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CHAPTER 1: INTRODUCTION

The objective of this report is to highlight all the resources and information collected, along with the preliminary steps and processes taken in building a word building game, WordEureka!.

1.1 Project Template

The project template chosen for WordEureka! is Project Idea Title 2 – Word-Based Game (CM3050 Mobile Development).

1.2 Project Concept

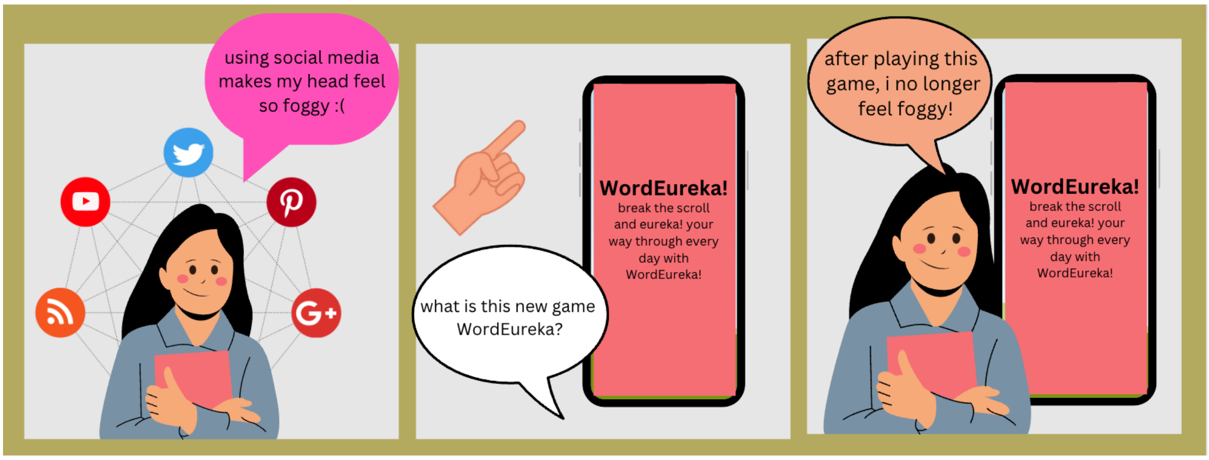


Figure 1 - WordEureka!'s storyboard

A commonly faced issue in youngsters today is that of 'brainrot' [19], in which individuals experience symptoms of mental fogginess, poor memory, and cognitive decline amongst others . Professionals believe that brainrot happens to be a side effect of increased social media usage. With mobile devices being highly accessible, social media becomes an aspect of life that is unavoidable.

WordEureka! is a word building game that hopes to challenge the brain and help improve mental health by providing an alternative to incessant scrolling on social media. It targets Singaporean youths of ages 13 - 25 as they are the primary users of many social media platforms. On top of being a word building game, it would include a 'word of the day' function to help users better their vocabulary, a friend mode so individuals can play against friends, customisable settings, and game statistics to track the user's progress. Additionally, WordEureka! is secure as it does not retrieve or store sensitive data like banking details, gender, or age.

1.3 Aims and Objectives

The aim of WordEureka! is as follows:

- To combat "brainrot" in Singaporean youths (aged 13-25) by providing an accessible and useful alternative to excessive social media usage, thereby bettering their mental health.

The objectives of WordEureka! are as follows:

- To provide an engaging mechanism behind the word building game that will promote cognitive skills and thus help prevent brain rot
- To develop social features to promote real life social interaction over social media interaction

- To develop customisable application settings to make the application accessible to all

1.4 Deliverables

The deliverables for this project can be split into two categories - the main deliverable and the assignment deliverables.

The main deliverable will be the application itself. Named WordEureka!, it is a word building game application for mobile devices.

The assignment deliverables include the preliminary report, feature prototype, and demo video for the midterm submission. The assignment deliverables for the final submission includes the final report, the final working prototype, and a presentation video.

Chapter 1 Word Count: 380/1000

CHAPTER 2: LITERATURE REVIEW

This section of the report begins by discussing research and studies that explore and justify the claims of WordEureka!. This section will then demonstrate gaps that WordEureka! will attempt to fill by exploring pre-existing applications in the same domain.

2.1 Brain rot and how it manifests

An article by Newport Institute defined brain rot as a syndrome caused by excessive screen usage which manifests as sluggishness, shortened attention span, and cognitive decline [19], and can have a multitude of negative effects on the mental health of youngsters such as difficulties in retaining and classifying information, decision-making, and problem-solving.

The article discussed how incessant scrolling on social media can turn into a form of addiction as it increases dopamine, the “feel-good” hormone [19]. This results in the brain associating scrolling with feelings of happiness and fulfilment even when the individual is conscious of the detrimental effects it may carry. Their research showed that the negative effects on attention and memory can even manifest physically over time as it might have changed the brain’s grey matter. Brainrot could manifest in other forms such as doom scrolling, where an individual actively seeks out disturbing information and adverse news, or zombie scrolling, where a user makes a habit of mindlessly scrolling with no real benefit [19].

It also discussed multiple ways such as limiting time on social media or doing a digital detox for youngsters to battle brain rot and its side effects, and highlighted playing word games [19] as a method to challenge the brain and keep it active, and to grow it with exertion over time . This supports WordEureka! in achieving its aim to help youths improve their mental health by providing a better alternative to incessant scrolling - a word building game.

2.2 Social media prevalence in Singapore and its negative impact on youths

Meltwater, a company that monitors online media, highlighted the high prevalence of social media in Singapore in 2024. It reported that there were 5.13 million identities on social media out of the overall population of 6.03 million, indicating an 85% penetration rate [21] (Figure A1, Appendix A). It further revealed that out of the daily average of 4 hours and 31 minutes that individuals spent on their phones, a significant 29.5% of it was spent on social media platforms with only 15.8% on productivity apps [21] (Figure A2, Appendix A). It also noted how Singaporeans use a monthly average of 6.9 social media platforms (Figure A3, Appendix A).

With the high presence of social media in Singapore came the necessity to look out for its potentially harmful side effects. During a sitting at parliament in February 2024 [12], Singapore’s Prime Minister Lawrence Wong addressed mental well-being as a crucial priority for the nation as youngsters of the current generation were becoming increasingly concerned about their mental health. The Straits Times reported that he highlighted how there are multiple factors stemming from social media that could negatively impact mental health, including but not limited to the urge to continuously project a favourable online identity, cyber bullying, and FOMO (fear of missing out) [12]. Additionally, he noted that increased social media usage could inhibit the development of a healthy brain as it results in decreased physical activities, lack of sleep, and reduced face-to-face communications. This eventually leads to youngsters struggling with the fundamental intellectual and emotional abilities required to lead a healthy adult life, making it crucial to cultivate good social media habits as early as possible.

These statistics highlighted the growing concerns surrounding the negative impacts of social media on mental health, especially among Singaporean youths. While it may be impossible to fully eradicate the usage of social media due to its high prevalence, a word building game such as WordEureka! could serve as a beneficial alternative to incessant scrolling. Word-building challenges, which lie at the core of WordEureka!, are able to prompt mental activities like language enhancement, memory recollection, and critical thinking abilities. Thus, this word building game could potentially mitigate the adverse effects of social media, improve their mental health, enhance vocabulary development, and heighten intellectual abilities such as memory function and problem-solving skills.

2.3 Brain teaser games effects on cognitive skills

A journal published by the Iranian Neuroscience Society on Basic and Clinical Neuroscience (BCN) [13] concluded that there was a positive correlation between playing brain teaser games and cognitive behaviours.

The journal began by explaining the two types of stress generated in people playing brain teaser games – limit stress and logic stress – both of which had a positive impact on cognitive abilities such as attention, problem-solving and concentration [13]. Logic stress was observed in puzzle games where there are no time limits, while limit stress was observed in games where the player could lose (time stress). These improved cognitive abilities, hand-eye coordination, attention, and concentration due to their effects on the frontal lobe of the brain [13] .

In this study, participants were required to play a brain teaser game and cognitive tests were performed on them using PASAT software (used to assess information and cognitive processing speed) [5] before and after the game in order to examine their cognitive abilities. The test's results (Figure B1, Appendix B) showed how the experimental group showed a significant improvement in attention after playing the game while there was little to no difference in the control group who did not play the game. The test revealed an increase in the stress levels of the experimental group after the game (Figure B2, Appendix B), while there was little to no difference in the control group [13].

Based on this research, incorporating time constraints in a word game could enhance mental abilities and health. This finding informed the design of WordEureka!, suggesting that integrating time limits in the game could lead to better cognitive abilities and improved mental health as it combines elements of limit and logic stress, similar to the game used in this study by the Iranian Neuroscience Society. The exploration of this study is crucial as it provides a scientific rationale for the development of this application.

2.4 Pre-existing Games

This section of the literature review focuses on pre-existing word based games currently in the market, namely Wordle, Wordscapes, and Spelltower [4]. These games were analysed based on the following metrics – game mechanics, user interface, and user reviews. Reviewing the benefits and drawbacks of these games helps in the development of WordEureka! as similar strategies to what makes them successful could be implemented, and the problems these pre-existing games face could be avoided. This review also allows for the identification of gaps in the market, and understanding the target audience and tailor WordEureka! to suit them.

2.4.1 Wordle

Figure 2 - Logo of Wordle

A word game that took the world by storm was Wordle [22], where each player gets six chances to guess the five-letter word of the day. For every guess, the colour of the letter's tile changes to green (if the correct letter is in the correct position), yellow (if the letter is in the word, but in a different position) or grey (if the letter does not belong in the word). With an average rating of 4.5/5 on Google, Wordle has an interesting gameplay and is highly accessible – colour blind users can change its colour scheme to one that is more visible for them [22]. The game went viral due to its exclusivity as users can only play it once every 24 hours, and because users on Twitter shared their scores in the form of emojis (Figure 3) without revealing the answer. An article by The Indian Express that discussed Wordle claimed that its success is because it is easy to play, exclusive, free, and accessible [22].

Figure 3 - A tweet by the creator of Wordle

An article by the NYPost [9] also discussed some of the drawbacks of Wordle. This included major backlash from British players as Wordle used the American spelling of words [9] and how it did not offer a multiplayer mode, restricting it to solo gameplay only [9], thereby potentially limiting long term engagement with the game and promoting solitude. The article also discussed how Wordle could possibly be detrimental to users' mental health when they fail to guess the right word and get frustrated [9].

From a technical perspective, Wordle fairs excellently in terms of user interface, game mechanics, and user reviews. While it has good aspects such as boosting memory and concentrating the mind, it shows that even though it became popular because of its exclusivity, a broader approach is required to guarantee its long term success.

WordEureka!, which takes its inspiration from Wordle, attempts to appeal to a wider audience by offering a simple gameplay experience and an easy-to-use UI. By learning from Wordle's limitations, WordEureka! aims to provide a more engaging platform by exploring social features such as a multiplayer mode or a focus on daily progression rather than winning the game. WordEureka!'s goals of fostering user interaction and long-term engagement may be supported by these considerations which are aimed at creating a more engaging and well-rounded gaming experience.

2.4.2 Wordscapes

Figure 4 - Logo of Wordscapes

Wordscapes is a free for all word game that allows users to test their vocabulary while having fun [20]. It has over 6,000 levels to unlock and is available on both iOS and Android. A mix of crossword and word search puzzles, it boasts a rating of 4.4/5 on Google Play. The game has features such as global and local leaderboards fostering a sense of community, and daily puzzles that, albeit, can only be unlocked after a user achieves a certain level in the game [20].

However, some aspects of the game deterred users. Even though Wordscapes was free to play, it came with a lot of advertisements that users found invasive and distracting from the actual game [3]. Users reported that the game tends to be bugged and did not give them the rewards like it was supposed to. The game, though free, hosts certain paid features like the piggy bank [3]. The sound effects and animations used in the game were reported to lag and make the game feel clunky and glitchy.

Inspired by Wordscapes, WordEureka! considers enhancing user experience by potentially integrating multi-level gameplay and diverse play styles. WordEureka! could also explore features like leaderboard and statistics to foster a sense of community, and an ad-free experience.

2.4.3 SpellTower

Figure 5 - Logo of SpellTower

SpellTower, a word puzzle game available on both iOS and Android, is a game in which players must clear the screen by connecting letter tiles to make words before the screen overflows with tiles. It offers different modes of gameplay such as Daily Search, Daily Tower, Tower, Rush, and Puzzle [2] (Figure 6). This gives users different choices and attracts them, allowing them to play for long by keeping them entertained and intrigued.

Figure 6 - Different play modes

According to CNET [15}, the game's success could be attributed to the fact that it is a combination of two famous games, namely Tetris and Boggle. The various gameplay modes, multiplayer, and battle modes have led to its success. The simple yet attractive and colourful user interface paired with the changeable colour schemes and customisable sound effects act as the cherry on top of the cake that is SpellTower [15]. A game of both vocabulary and strategy, SpellTower was named among IGN's most underrated games [16] .

However, despite its roaring initial success, the game currently stands at only 3/5 stars on Apple's App Store due to the excessive amounts of advertisements users who have not paid to unlock the full version of the game receive. Furthermore, certain elements of the game such as bonus game modes and detailed statistics are only available in the fully paid version of SpellTower (Figure 7).

Figure 7 - Features of SpellTower which have to be paid for to be unlocked

WordEureka!, learning from the success of Spelltower, plans to implement engaging gameplay to attract users, and additional features such as game statistics and leaderboards. WordEureka! aims to offer different gameplay modes to keep the game engaging, similar to SpellTower. As opposed to Spelltower, WordEureka! aims to provide all features to all users since the ultimate aim of the application is to provide an accessible alternative to social media and help improve mental health. Additionally, WordEureka! will present an ad-free experience as opposed to Spelltower's ad riddled game interface.

2.4.4 Summary

Table 1 - Comparison of all applications

Application Name	Features								
	Does the app have accessible user interface settings?	Is the app free?	Does the app have premium features that are available only when paid for?	Does the application have invasive advertisements ?	Does the game have easy to understand gameplay?	Does the app have free additional features such as:			Rating
						leaderboard	game statistics	different game modes	
Wordle	✔	✔	✔	✘	✔	✘	✔	✘	4.6/5
Wordscapes	✘	✔	✔	✔	✔	✘	✔	✘	3.7/5
SpellTower	✘	✔	✔	✔	✔	✘	✘	✘	4.3/5
WordEureka!	✔	✔	✘	✘	✔	✔	✔	✔	-

By analysing the strengths and weaknesses of popular word building games, WordEureka! is able to carve its own niche in the market. The analysis revealed the following factors that contribute to the success of a word based game:

1. Accessible UI - Wordle demonstrated the importance of accessibility in a game application
2. Engaging Gameplay- SpellTower showcases the significance of an interesting game mechanism in contributing to a game's success
3. Social Features - Wordscapes social features like game statistics are one of the factors that contributed to it success

Based on these insights, WordEureka! aims to provide a well-rounded platform through its social features, ad-free experience, and interesting gameplay as highlighted in Table 1. Building upon the best features of pre-existing games and by improving upon the undesirable features of those games, WordEureka! aims to create a word-building game that stands out from the competition and provides a satisfying and enjoyable experience for its user.

2.5 Conclusion

This literature review discusses the effects of incessant social media usage, also known as "brain rot," and its impact on Singaporean youths' mental health. Research suggests that cognitive stimulation through brain teaser games can help counteract these negative effects. The review also analyses popular word games like Wordle, Wordscapes, and SpellTower, identifying their strengths and limitations.

WordEureka! is proposed as a solution to these issues, combining cognitive stimulation with an ad-free, engaging user experience, designed to offer a healthier alternative to social media while improving mental health.

Chapter 2 Word Count: 2344/2500

CHAPTER 3: PROJECT DESIGN

This section of the report discusses the design and development processes behind WordEureka!. It dives into the key aspects of the project such as the domain, design choices, application architecture, tools used, the work plan, and the testing and evaluation methods to be used.

3.1 Domain and Users

WordEureka!'s target audience is Singaporean youths of ages 13 - 25 as they are more susceptible to the negative side effects of prolonged social media usage. According to the American Psychology Association (APA) [34], the adolescent brain desires for attention and becomes more sensitive to comments from peers. This makes them highly engaged with social media platforms and digital content, thus making them vulnerable to the harms of social media.

However, although the primary target audience of the game is youths, anyone who is interested in such games are considered secondary users of WordEureka! This application falls under the domain of educational entertainment and casual learning as it aims to provide an engaging experience while stimulating cognitive skills.

3.2 Design Choices

3.2.1 Software Development Lifecycle (SDLC) Method

The SDLC method chosen to guide along the process of WordEureka! is the Agile Software Development Lifecycle [14]. Mobile app development often involves frequent updates, new features, and continuous improvement through regular user feedback. This calls for a flexible and adaptable work style, making the Agile SDLC the most suitable.

The five stages of the Agile SDLC are ideation (brainstorming and gathering user requirements), development (coding and creating the app's features), testing (ensuring functionality and fixing bugs), deployment (releasing app to users), and operations (monitoring and updating based on user feedback)[14].

3.2.2 Colour Scheme

As part of the development stage of WordEureka!, the colour scheme for the application's interface was decided first. A shade of red (#CA0800) was chosen as the primary colour for the application due to its ability to work as a call to action [17], thus attracting users to play WordEureka!. The Canva Colour Wheel [10] was then used to create a highly contrasting yet balanced tetradic palette to be used as the primary colour scheme based off #CA0800. Additionally, since the application aims to be accessible to users who may be visually impaired or colour blind, Venngage, a professional infographic creating tool [24], was used to come up with an accessible colour scheme based on the primary colour scheme. Both colour schemes can be seen in Figure 8. The application will be set to the primary colour scheme on default, but will allow users to switch to the accessible colour scheme if they turn on 'high contrast mode' in the game's settings.

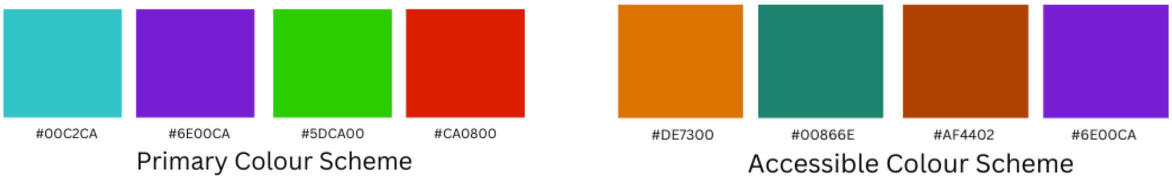


Figure 8 - Colour schemes

3.2.3 User Flow Diagram

A user flow diagram is important in understanding how users would interact with WordEureka!. Outlining the different routes a user can take facilitates navigation and ensures a seamless user experience. This is crucial for WOrdEureka! since user-friendliness is crucial to maintaining interest. A well designed user flow diagram would facilitate the identification of potential bottlenecks, improve user satisfaction, and direct design choices that correspond with user preferences and behaviours [1].

Figure 9 shows the user flow diagram of WordEureka!. This diagram was built centred around the user's needs and behaviours and shows the core features and the main navigation. The diagram has a clear entry point and starts at the homepage of WordEureka! and has 5 branches - previous games' statistics, settings, game rules, word of the day, and play game. When the user picks one of the first 4 options, it brings them to the respective pages, and an option to go back to the homepage. When the user chooses the play game option, they are able to choose between multiplayer or solo mode, both of which bring the users through the process of playing the game with options to quit, pause, or restart the game during the gameplay, and to share scores or replay game when the game is over.

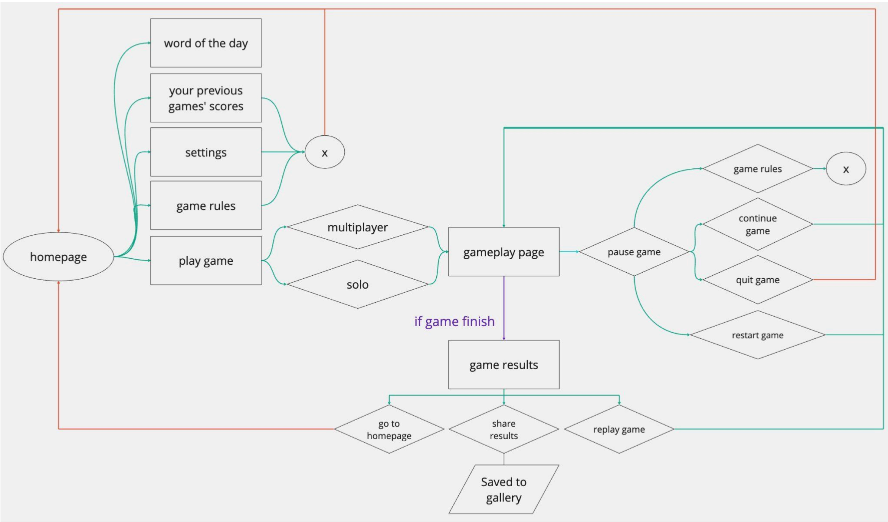


Figure 9 - User flow diagram

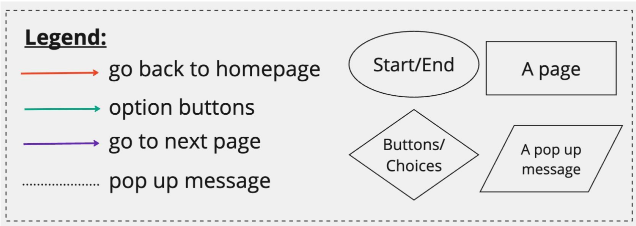


Figure 10 - Legend for user flow diagram

3.2.4 Wireframe

User-centred design principles were the main focus of the wireframe creation process of WordEureka!. Key aspects included intuitive experience, usability, accessibility, and visual hierarchy. Wireframes also allow for the identification of any usability issues early on, and to confirm the user experience before developing the application itself [11]. Developed from the low fidelity version to the high fidelity version over time, the wireframes allow space to incorporate accessibility and attractability during the coding process. Figure 11 shows the high fidelity wireframe of WordEureka! in light mode and light & high contrast mode. Figure C1 and C2 in Appendix C show the low and mid fidelity wireframes respectively, while Figure C, shows the high fidelity wireframe of WordEureka! in the dark mode and dark & high contrast mode.the step by step creation of the wireframes from low to high fidelity helped ensure that the final design satisfies user needs and is in line with the app's objectives.

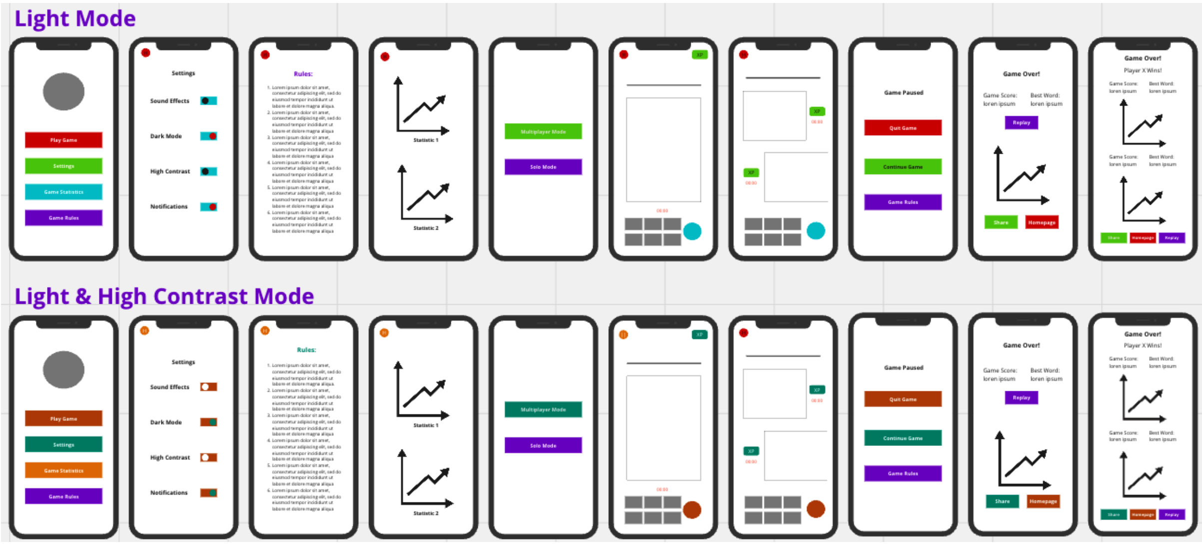


Figure 11 - High fidelity wireframe in *Light Mode* and *Light & High Contrast Mode*

3.3 Project & Application Structure

The project architecture of WordEureka! will follow that of a simplified mobile application architecture [23]. The user will interact with the application on the presentation layer which is responsible for the user interface and presentation logic, while the application's data layer will be interacting with the APIs implemented in order to support the functionality of the game. It aims to access the device's local storage through AsyncStorage in order to store data such as game scores. The application structure of WordEureka can be seen below in Figure 12.

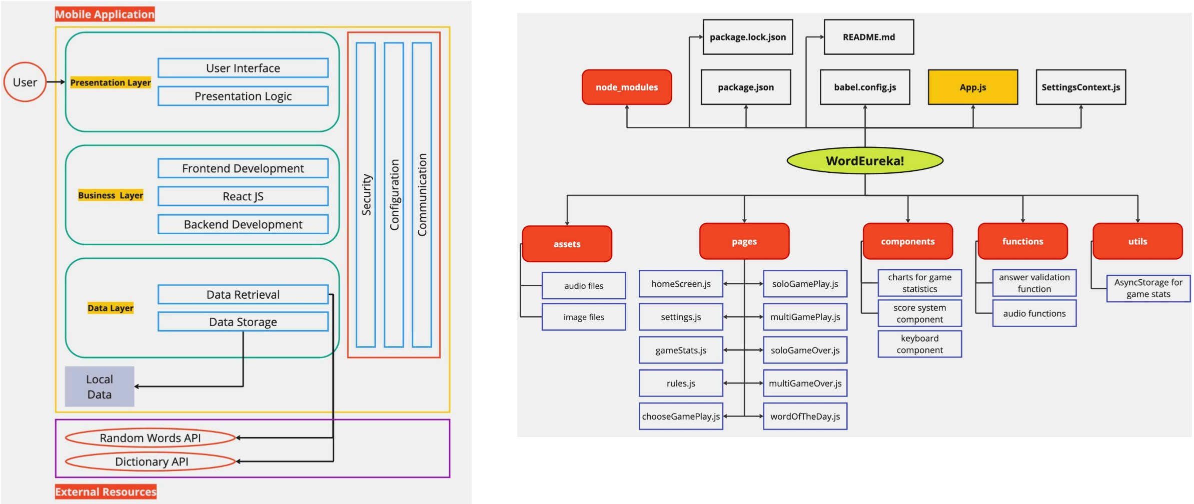


Figure 12 - Mobile application structure and WordEureka! project structure

The project structure consists of the necessary files and folders to run WordEureka!. It is designed for clarity, modularity, and scalability. Organising the files into folders such as assets, pages, components, functions and utils promotes separation of concerns. This ensures that every part of the application can be developed and tested independently. Additionally, the usage of a central App.js file streamlines the development process by managing the overall UI.

3.4 Tools To Be Used

- React Native Framework [32]: The React Native framework will be used to code this application. Using React Native allows the use of same code for both iOS and Android platforms, making WordEureka! accessible to as many users as possible. This increases efficiency in the process of app development.
- GitHub [31]: GitHub will be used to store and manage the code. It allows for easier version and access control, and to keep track of tasks, resulting in better project management.
- Expo Snacks [25]: Expo, an open-source framework for building cross-platform applications, will be used to code WordEureka!. Expo allows for immediate live previews of the app on Android, iOS, and the web, facilitating rapid development and testing.
- APIs
    - The first API will be the Wordnik Random Word API [33] which will be used to retrieve random words for the gameplay. Its extensive word database and real-time updates make it ideal for generating diverse and engaging content for the game.
    - The second API will be the Wordnik Word Of The Day API [33] which will be used to retrieve the word of the day and its definitions.
    - The third API will be the Merriam Webster Collegiate Dictionary API [28] which will be used to check and validate the user's inputs against the dictionary during gameplay. Being a reputable source, it gives players dependable feedback and enhances both their gaming and educational experience.
- Pixabay [36] and Freesound [30]: Additionally, sound effects will be added to the game in order to create a more enticing and engaging atmosphere for the players. Resources will be retrieved from these websites as they offer royalty free audios.

3.5 Work Plan

As part of the development stage in the Agile SDLC [14], a Gantt Chart to outline the tasks of building WordEureka! over the next few months is portrayed in Figure 13. The bulk of the time leading up to June 17th will be spent on building a feature prototype and on the preliminary report.

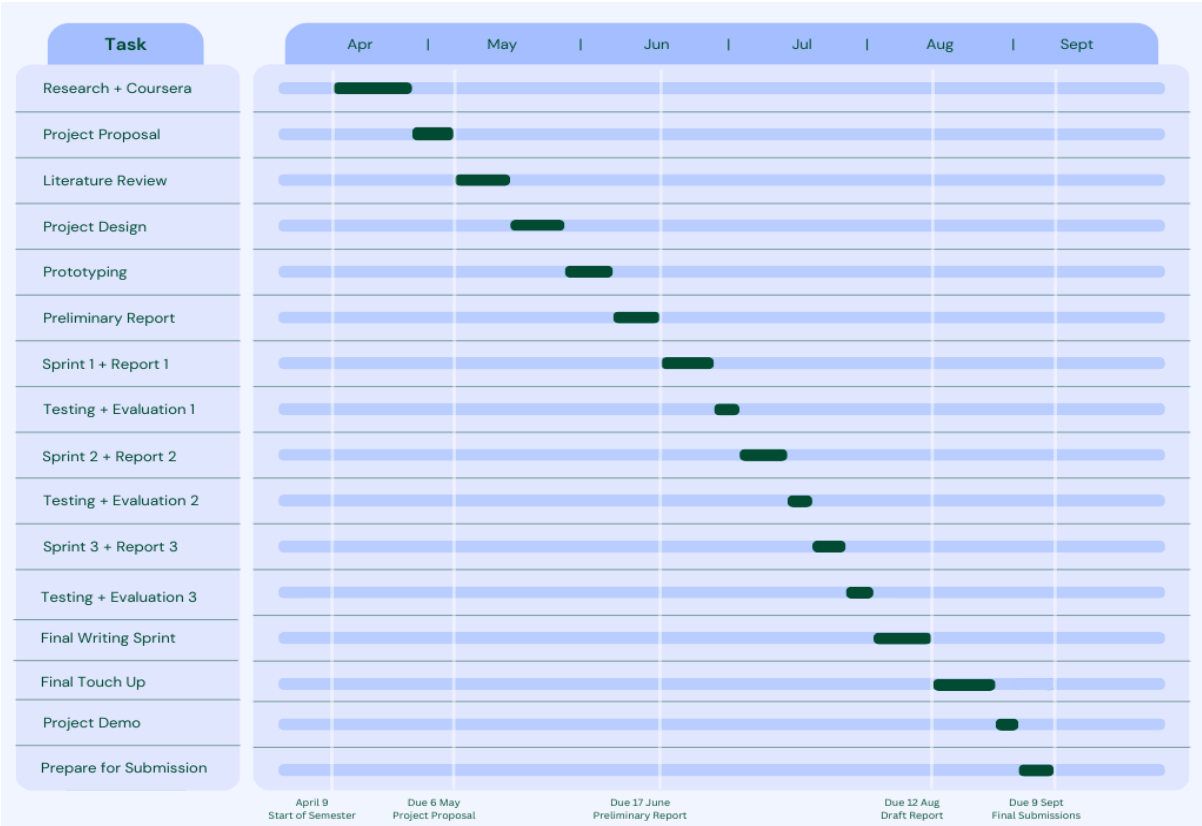


Figure 13 - Gantt Chart for WordEureka!

The first ten days after June 17th will be spent on the first coding sprint. This time will be focused on working on the backend components of WordEureka! such as implementing the social features while simultaneously writing the final report. After this, the first testing session will be used to test and evaluate the backend features implemented.

The next ten days will be spent on the second coding sprint which will be used to implement any changes based on the feedback received from the first testing period, and on the frontend features of the application. The final report will be simultaneously written up over this time. Three days after this will be spent on the second testing and evaluation session which will be used to gather feedback on the frontend features implemented, and the improved backend features.

A final period of one week will be spent on coding any unimplemented features of the application, along with working on the feedback received over the past testing sessions. Upon reaching a state of near-completion (approximately 95%), a final round of user testing will be used to gather any feedback on this version of WordEureka!. The week leading up to the submission of the draft report will be spent on any code refinement and on polishing the report based on feedback.

After the draft report submission on August 12th, the 4 weeks before the final submission will be spent on completing any other essential coding tasks for the final version of the application, refining the written report based on any additional feedback, and preparing a high-quality demo video for submission, showcasing WordEureka!'s features.

Dividing the workload of WordEureka! in this structured approach allows for enough flexibility to work on different aspects of the application while maintaining work discipline and ensuring steady progress towards the final application.

3.6 Testing and Evaluation

Moving into the third stage (testing stage) of the Agile SDLC, a two step approach will be implemented in order to test and evaluate the success of WordEureka!.

3.6.1 Testing

There are three ways in which WordEureka! will be tested:

1. Unit Testing (white box testing)
    a. Will be conducted using manual testing [27]
    b. Ensures that every individual component and function in the application is working as expected according to the test cases listed in Appendix E
2. Integration Testing (white box testing)
    a. Will also be conducted through manual testing
    b. Ensures that different parts of the application work together as expected in accordance with the test cases listen under Appendix E
3. User Testing (black box testing)
    a. The System Usability Scale (SUS) [18] will be used in the user testing sessions of WordEureka!
    b. Found in Appendix E, this scale will be used to gather user inputs on the final iteration of WordEureka! as it is a reliable tool and focuses on overall usability
    c. Feedback will be gathered on the usability aspects of the application.

Manual testing will be used to test the application because it provides a comprehensive analysis of all the components and how they interact with each other. It also gives insights into a user's perspective on how the application works, allowing for a nuanced understanding of the app's usability and user experience. A combination of white box and black box testing is crucial as it allows for feedback from both a developer and a user perspective. The feedback received through testing would result in the final version of the application to be as perfect as possible, and to cater to as many people. All surveys for user testing will be conducted through Google Forms in order to facilitate the consolidation of information.

3.6.2 Evaluation

Similar to user testing, a variety of potential users will use WordEureka! when it is nearing completion, it will be evaluated against the project objectives highlighted in chapter 1, by asking users to rate the statements found in Appendix E on a scale of 1-5, with 1 being strongly disagree and 5 being strongly agree.

Additionally, interviews with users to gain a deeper understanding of users' thoughts and feelings while playing WordEureka! will be conducted. During this interview, a modified version of the

Perceived Stress Scale (PSS) [6] (a tool used to assess an individual's perceived stress levels) will be used to evaluate a user's stress levels before and after playing WordEureka!. Users will rate statements on a scale of 0 (never) - 4 (very often) and their responses will be totalled up giving them a score out of 40. The higher their score, the higher their perceived stress.

The lower the PSS score after playing the game, the better the application is at contributing to users' positive stress. The PSS allows for a reliable and straightforward understanding of how different users react while playing WordEureka!. The modified questions will ask the users about their feelings and thoughts whilst playing the game. Conducting evaluations will ensure that WordEureka! is of the best quality, and that the users' needs are met. Therefore, it is an essential step in the production of this application.

Chapter 3 Word Count: 2186/2000

CHAPTER 4: IMPLEMENTATION

This chapter details the process of developing and improving WordEureka!, with an emphasis on the iterative method used to improve the functionality and user interface of the application. This section also discusses technical implementations like AsyncStorage, stack navigation and API interactions, and design aspects like sound effects integration and responsive UI.

4.1 Iteration 1 and Feedback on App Features

During the initial stages of development, potential users were provided with a bare bones application with the following key features implemented and were asked for their feedback on them:

- Word of the day feature: To help improve vocabulary
- Shareable score: To promote social interaction
- Past game statistics: To track progress across games
- Solo and multiplayer options: To offer flexibility in gameplay.
- Customisable settings: For personalised and accessible user experience.
- Engaging sound effects and animations: To improve user engagement.
- Push notifications: To keep users informed and engaged.

The feedback results are summarised in figure 14 and table 2 below.

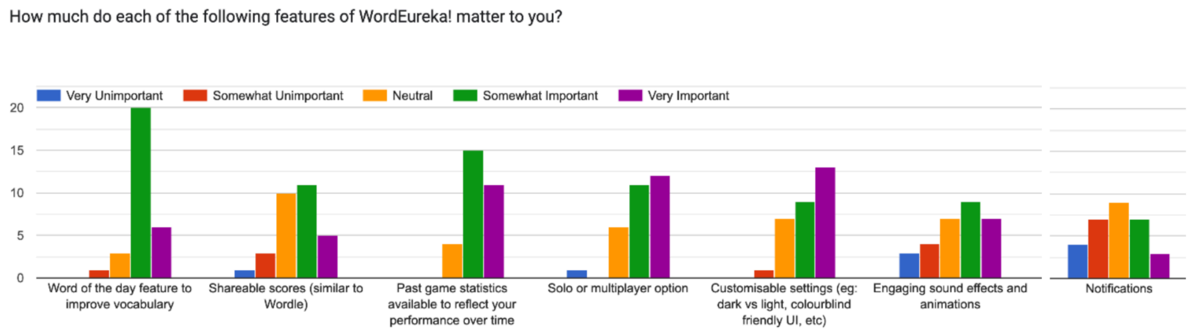


Figure 14 - Bar chart on user survey

Feature	Percentage of respondents that chose somewhat/very important (%)	Percentage of respondents that chose neutral (%)	Percentage of respondents that chose somewhat/very unimportant (%)
Word of the day feature to help improve vocabulary	86.7	10	3.3
Shareable score to promote social interaction	53.3	33.4	13.3
Past game statistics to track progress across games	90	0	10
Solo and multiplayer options	76.7	20	3.3
Customisable settings	73.3	20	3.3
Engaging sound effects and animations	53.3	23.4	23.3
Push notifications	33.3	30	36.7

Table 2 - User survey data statistics

As shown in the diagrams above, out of the main features that WordEureka! had planned to implement, the users felt that the push notifications were not necessary in the game as 36.7% voted for “somewhat” or “very unimportant” and 30% voted for neutral. Consequently, it was decided to exclude push notifications from the final version of WordEureka! based on this input.

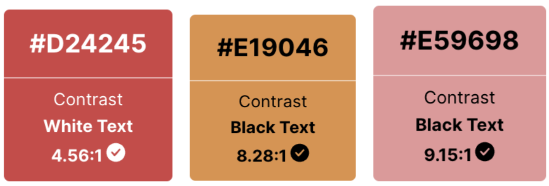
Conversely, there was an overwhelming percentage of users that voted for “somewhat” or “very important” for the other features.These features will therefore be prioritised and incorporated into the final product to meet user expectations and enhance the overall gaming experience in WordEureka!.

Since WordEureka! is a user centric application that aims to better the mental health of users, this feedback was instrumental in the development process of WordEureka as it helped ensure that the final product would align with the users’ needs and expectations. It also essentially helps support users’ wellbeing by prioritising features that were found valuable and by discarding features that were deemed unnecessary.

4.2 Iteration 2 and Feedback

After receiving feedback on the app's features in the first iteration, the second coding sprint focused on refining essential features, discarding unnecessary ones, and implementing UI elements such as colours and layout based on the high-fidelity wireframes. Once this iteration of WordEureka! was completed, users were asked for feedback, this time on the UI design rather than the app's features.

Users provided input on the layout, element placement, and colour scheme across various pages, as well as suggestions for improvement. The most notable feedback was that the initial colour scheme (Figure 8) was too jarring and incoherent. In response, a new, more accessible color palette was developed using Canva and Venngage, as shown in Table 3.



accessible colour palette

Original colour palette

Table 3 - Colour palette changes

Additionally, users felt the background appeared too plain and boring. To address this, a simple yet engaging wallpaper was created and integrated into the pages, making the design more vibrant and attractive. Many respondents also suggested rounding the button corners for a more polished look. This feedback was critical, as WordEureka! is a user-centric app, and ensuring it meets user expectations is essential. Figure D1(Appendix) summarises the feedback, while Table D1 (Appendix) illustrates the UI changes implemented based on this input.

4.3 Final Iteration

The main features of WordEureka are as follows - solo gameplay, multiplayer gameplay, word-of-the-day (WOTD), previous game statistics, customisable settings, and game rules.

The users of wordeureka can choose between playing a game on their own (solo gameplay), or they can play the game with their friend. (multiplayer game play). When they start the game, the application retrieves a word and starts the one minute countdown to let the player play the game. In game mode, the user can interact with the application by entering their words with a keyboard, and pausing the game which will allow them to restart, resume or quit the game, and refer to the rules if they require it. When the game ends, they can also replay the game, share their game scores or go back to the homepage of the application.

Apart from playing the game, users can also learn something new everyday through the WOTD feature. It gives users the definition to a new word everyday, and sample sentences so users can learn how to use them. This allows them to exercise their brains and keeping it active by increasing their knowledge. There is also a page with the game rules that will allow the users to understand the game properly before playing, and will let them refer back to it whenever they need to. They also have customisable settings for dark mode, high contrast mode, sound effects, and notifications so users can find the most suitable interface and user experience for themselves. Additionally, users will also be able to track their progress over all the games they play as the game stores information about the games they play such as score, timestamp, and the word with the highest score.

4.3.1 Technical Implementations

4.3.1.1 API Retrievals

One of the main functions behind WordEureka! is API retrieval. figure 15 below depicts how wordeureka interacts with the three different APIs mentioned in Chapter 3.

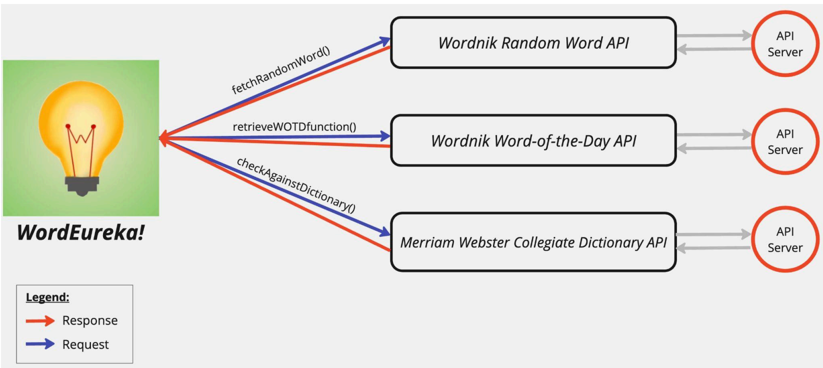


Figure 15 - Graphical representation of API Retrivals

In this scenario the client would be wordeureka. Different components of the application interact with different APIs in order to allow wordeureka to work seamlessly.

Sologameplay.js and multigameplay.js both use fetchRandomWord() to fetch a random word from the Wordnik API ito start the game. This asynchronous function shown below in figure 16 sends a HTTP GET request to the API and parses the JSON response from the API and assigns the random word retrieved to variable ‘word’. The function then checks that this word consists only of alphabetic characters before converting it to uppercase and assigns it to ‘randomWord’.

```
export const fetchRandomWord = async ( wordsArray, setRandomWord, startingLetter, endingLetter, setGameEnded, setTimeRemaining,
setTimerActive,  setTimerPaused,setGameID, inputRef
) => {
  try {
    const response = await fetch(
      'https://api.wordnik.com/v4/words.json/randomWord?api_key=bw6jvz79rhbznnfide5n25x9eo6box9nx2ex74h92euv0k0us'
    );
    const data = await response.json();
    let word = data.word;

    if (/^[A-Za-z]+$/.test(word)) {
      word = word.toUpperCase();
      wordsArray.push(word);
      setRandomWord(word);

      const starting = word.charAt(0);
      startingLetter.push(starting);
      const ending = word.charAt(word.length - 1);
      endingLetter.push(ending);

      setGameEnded(false);
      setTimeRemaining(60);
      setTimerActive(true);
      setTimerPaused(false);

      const newGameID = uuidv4();
      setGameID(newGameID);

      console.log('SOLO GAME STARTED:', newGameID);
      console.log('Random word retrieved:', word);

      inputRef.current.focus();
    } else {
      console.log('Word wrongly fetched:', word);
      fetchRandomWord(wordsArray, setRandomWord, startingLetter, endingLetter, setGameEnded, setTimeRemaining, setTimerActive,
setTimerPaused, setGameID, inputRef);
    }
  } catch (error) {
    console.error('Error fetching random word:', error);
  }
};
```

Figure 16 - fetch random word API Code function

During gameplay when a user submits their word,, the asynchronous function checkAndValidateWord () shown below in figure 17 is triggered. This function takes the submitted word as ‘userSubmittedWord’ and sends a HTTP GET request to the Merriam-Webster API using the fetch function by including userSubmittedWord as part of the URL. It then checks if the API returns an array with a valid uuid. If the uuid exists, the user is awarded points for the word. If not, the word is rejected and the player is not awarded any points.

```
const checkAndValidateWord = async () => {
 if (startingLetter.length > 1 && endingLetter.length > 1) {
   const currentWordStartingLetter = startingLetter[startingLetter.length - 1];
   const prevWordEndingLetter = endingLetter[endingLetter.length - 2];

   //if letters match, move onto next step
   if (currentWordStartingLetter === prevWordEndingLetter)
   {
     const userSubmittedWord = wordsArray[wordsArray.length - 1];
     try
     {
       const response = await fetch(`https://dictionaryapi.com/api/v3/references/collegiate/json/
       ${userSubmittedWord}?key=6598d205-4db2-46dd-8f0b-f488f7ccc137`);
       const data = await response.json();

       if (Array.isArray(data) && data.length > 0)
       {
         const firstEntry = data[0];
         // word is accepted if it has a uuid
         if (firstEntry && firstEntry.meta && firstEntry.meta.uuid)
         {
           const uuid = firstEntry.meta.uuid;
           calculateScore(userSubmittedWord);
           totalGameScore();
           setUserInput('');
           inputRef.current.focus();

           // Generate a random number between 0 and 1
           const randomChance = Math.random();
           // Set a threshold for triggering the feedback (e.g., 0.3 gives a 30% chance)
           const feedbackThreshold = 0.4;
           if (randomChance < feedbackThreshold)
           {…
           }
         }
         // word is not accepted if it doesnt have a uuid
         else
         {
           console.log(userSubmittedWord,'has no UUID, so it is not accepted');
           triggerCharacterFeedback('Invalid word. Try again!');
           startingLetter.pop();
           endingLetter.pop();
           wordsArray.pop();
           setUserInput('');
           inputRef.current.focus();
         }
       }
       else
       {
         console.log(userSubmittedWord,'has no data available, so it is invalid');
         triggerCharacterFeedback('Invalid word. Try again!');
         startingLetter.pop();
         endingLetter.pop();
         wordsArray.pop();
         setUserInput('');
         inputRef.current.focus();
       }
     }
     catch (error)
     {
       console.error('Error checking word validity:', error);
       triggerCharacterFeedback('Invalid word. Try again!');
       startingLetter.pop();
       endingLetter.pop();
       wordsArray.pop();
       setUserInput('');
       inputRef.current.focus();
     }
   }
   // immediate reject if letters dont match
   else {
     setUserInput('');
     inputRef.current.focus();
     console.log('letters do not match!');
     triggerCharacterFeedback('letters do not match :( ');
     startingLetter.pop();
     endingLetter.pop();
     wordsArray.pop();
     setUserInput('');
     inputRef.current.focus();
   }
 } else {
   console.log('Add more words first');
 }
```

Figure 17 - Word check and validation API Code function

wordOfTheDay.js triggers retrieveWOTDfunction() (in figure 18) through useEffect to interact with the Wordnik API. It sends a HTTP GET request to the API and the response is parsed and extracted to store the word, part of speech, definition, and examples. These are then displayed to the user on the WOTD page.

```
const retrieveWOTDfunction = async () => {
  try {
    const response = await fetch('https://api.wordnik.com/v4/words.json/
wordOfTheDay?api_key=bw6jvz79rhbznnfide5n25x9eo6box9nx2ex74h92euv0k0us'
    );
    const data = await response.json();
    const word = data.word;
    setWotd(word);
    const pos = data.definitions[0].partOfSpeech;
    setPartOfSpeech(pos);
    const wordDefinition = data.definitions.map((def) => def.text);
    setDefinition(wordDefinition);
    let wordExample = data.examples.map((ex) => ex.text);
    wordExample = wordExample.slice(0, 3);
    setExample(wordExample);
    console.log('WOTD retrieved successfully!');
  } catch (error) {
    setWotd('Loading...');
    setPartOfSpeech('Loading...');
    setDefinition(['Loading...']);
    setExample(['Loading...']);
    console.error('Error fetching word of the day:', error);
  }
};

useEffect(() => {
  retrieveWOTDfunction();
}, []);
```

Figure 18 - Word of the day API code function

4.3.1.2 AsyncStorage

AsyncStorage is an unencrypted and asynchronous storage system in React Native which is used to store data across app sessions [26]. Wordeureka uses asyncStorage to store the following information:

- game scores
- highest scoring word in each game
- Multiplayer win counts

Table 4 shows the pages that use functions to store/retrieve data from AsyncStorage. For example, sologameplay.js stores the score and highest scoring word of each game through storeGameScores() (Table 4) and addBestWordScore() (Table 4) while scoreLineChart.js retrieves this information through getGameScores() (Table 5) to plot a line graph showing the player's progress across games.

File	Functions to Store	Functions to Retrieve
../pages/sologameplay.js	• storeGameScores • addBestWordScore	
../pages/multigameplay.js	• addPlayerWin	
../pages/gameStats.js		• getBestWordScores • getGameScores • getPlayerWins
../components/pieChart.js		• getPlayerWins
../components/scoreLineChart.js		• getGameScores
../components/wordLineChart.js		• getBestWordScores

Table 4 - Table of file storage and retrieval

<table>
<tr>
<td><pre>export const storeGameScores = async (gameID, score) => {
  try {
    const scores = await AsyncStorage.getItem(GAME_SCORES_KEY);
    let gameScores = scores ? JSON.parse(scores) : [];

    gameScores.push({
      gameID,
      score,
      timestamp: new Date().toISOString(),
    });

    await AsyncStorage.setItem(GAME_SCORES_KEY, JSON.stringify(gameScores));
  } catch (error) {
    console.error('Error storing game score:', error);
  }
};</pre></td>
<td><pre>export const getGameScores = async () => {
  try {
    const scores = await AsyncStorage.getItem(GAME_SCORES_KEY);
    return scores ? JSON.parse(scores) : [];
  } catch (error) {
    console.error('Error retrieving game scores:', error);
    return [];
  }
};</pre></td>
<td><pre>export const addBestWordScore = async (gameID, word, score) => {
  try {
    const scores = await AsyncStorage.getItem(BEST_WORD_SCORES_KEY);
    let bestWordScores = scores ? JSON.parse(scores) : [];

    bestWordScores.push({ gameID, word, score });

    await AsyncStorage.setItem(BEST_WORD_SCORES_KEY, JSON.stringify
(bestWordScores));
  } catch (error) {
    console.error('Error storing best word score:', error);
  }
};</pre></td>
</tr>
<tr>
<td>storeGameScores()</td>
<td>getGameScores()</td>
<td>addBestWordScore()</td>
</tr>
</table>

Table 5 - Helper functions

Figure 19 below illustrates WordEureka! interaction with AsyncStorage where green arrows represent the actions of storing data and purple arrows show the process of retrieving data from AsyncStorage back into WordEureka!.

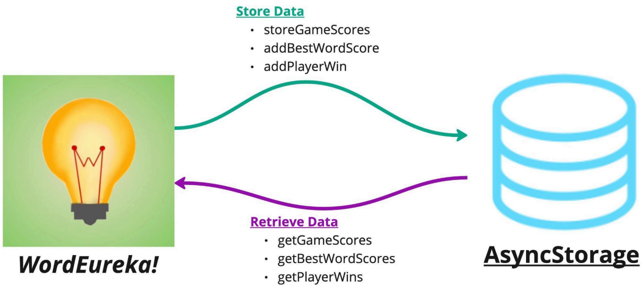


Figure 19 - App data storage visualisation

4.3.1.3 Stack Navigation

In wordeureka, stack navigation is essential in managing screen transitions [35]. It is implemented using createStackNavigator() in App.js, allowing for smooth transitions between the various pages in the application. The code in figure 20 shows the pages in the application such as HomeScreen, PlayGame, and GameRules. Each screen also has options set, like hiding the header and enabling the gestures. When the app is launched, the user is shown the HomeScreen as it is declared under initialRouteName. The user can then move to other pages of the application from here by clicking the respective TouchableOpacity. This approach is user-friendly as it mimics the flow of navigating through these pages and is effective as it provides a seamless and intuitive user experience.

```
// importing my pages
import HomeScreen from './pages/homeScreen';
import PlayGame from './pages/chooseGamePlay.js';
import SoloGame from './pages/soloGamePlay.js';
import MultiGame from './pages/multiGamePlay.js';
import GameRules from './pages/rules.js';
import Settings from './pages/settings.js';
import PastStatistics from './pages/gameStats.js';
import WOTD from './pages/wordOfTheDay.js';

const Stack = createStackNavigator();

export default function App()
{
  return (
  <SettingsProvider>
    <NavigationContainer>
      <Stack.Navigator initialRouteName="HomeScreen">
        <Stack.Screen name="HomeScreen" component={HomeScreen}
          options = {{headerShown: false, gestureEnabled: true,}}/>
        <Stack.Screen name="PlayGame" component={PlayGame}
          options = {{headerShown: false, gestureEnabled: true,}}/>
        <Stack.Screen name="GameRules" component={GameRules}
          options = {{headerShown: false, gestureEnabled: true,}}/>
        <Stack.Screen name="Settings" component={Settings}
          options = {{headerShown: false, gestureEnabled: true,}}/>
        <Stack.Screen name="PastStatistics" component={PastStatistics}
          options = {{headerShown: false, gestureEnabled: true,}}/>
        <Stack.Screen name="WOTD" component={WOTD}
          options = {{headerShown: false, gestureEnabled: true,}}/>
        <Stack.Screen name="SoloGame" component={SoloGame}
          options = {{headerShown: false, gestureEnabled: true,}}/>
        <Stack.Screen name="MultiGame" component={MultiGame}
          options = {{headerShown: false, gestureEnabled: true,}}/>
      </Stack.Navigator>
    </NavigationContainer>
  </SettingsProvider>
  );
}
```

Figure 20 - Page Imports and App Initialization

4.3.1.4 Scoring System

The scoring system in WordEureka! rewards players for their word-building skills. Table 6 shows the point each letter is assigned based on their frequency in the english language [29] (as shown in figure D2 in appendix D). For instance, the letter E which appears with the most frequency of 12.02 has been assigned 1 point, and the letters K, X, Q, J, and Z which appear with the least frequency are awarded 10 points. This scoring system might incentivize players to use less frequent letters, especially in multiplayer mode, in order to get more points. It also ensures that players can't achieve high scores too easily, making the game more challenging and rewarding.

Letter	Frequency	Point Assigned	Letter	Frequency	Point Assigned
E	12.02	1	M	2.61	8
T	9.1	1	F	2.3	8
A	8.12	2	Y	2.11	8
O	7.68	3	W	2.09	8
I	7.31	3	G	2.03	8
N	6.95	4	P	1.82	9
S	6.28	4	B	1.49	9
R	6.02	4	V	1.11	9
H	5.92	5	K	0.69	10
D	4.32	6	X	0.17	10
L	3.98	7	Q	0.11	10
U	2.88	8	J	0.1	10
C	2.71	8	Z	0.07	10

Table 6 - Scoring System letter weights

Figure 21 below shows the code in the file scoring.js. The object, scoresForLetters, acts as a dictionary that maps each letter to its respective point. The assignScore() function takes word as an input and matches the keys in scoresForLetters and iterates over each letter. If the letter exists in scoresForLetters, its corresponding score is added to the total score.

```
const scoresForLetters = {
    'A': 2, 'B': 9, 'C': 8, 'D': 6, 'E': 1, 'F': 8,
    'G': 8, 'H': 5, 'I': 3, 'J': 10, 'K': 10, 'L': 7,
    'M': 8, 'N': 4, 'O': 3, 'P': 9, 'Q': 10, 'R': 4, 'S': 4,
    'T': 1, 'U': 8, 'V': 9, 'W': 8, 'X': 10, 'Y': 8, 'Z': 10
};

function assignScore(word) {
    let score = 0;
    for (const letter of word.toUpperCase()) {
        if (scoresForLetters.hasOwnProperty(letter)) {
            score += scoresForLetters[letter];
        }
    }
    return score;
}
```

Figure 21 - Code Implementation of Letter Scoring

4.3.1.5 Sharing Results

Users can also share their game results by clicking the share button on the game results page. Figure 22 below shows the code that allows for this to happen and an example of the summary that is saved to the user’s device.

```
const shareResults = async () =>
{
  try
  {
    const { status } = await MediaLibrary.requestPermissionsAsync();
    if (status !== 'granted')
    {
      Alert.alert('Permission denied', 'Please allow access to your
photos.');
      return;
    }

    if (layoutReady)
    {
      const uri = await captureRef(viewRef, {
      format: 'png',
      quality: 0.8,
      });

      await MediaLibrary.saveToLibraryAsync(uri);
      Alert.alert('Success', 'Scores summary saved to your gallery!');
    }
    else
    {
      Alert.alert('Error', 'Layout not ready for screenshot.');
    }
  }
  catch (error)
  {
    console.error('Error capturing screenshot:', error);
    Alert.alert('Error', 'Failed to capture screenshot.');
  }
};
```

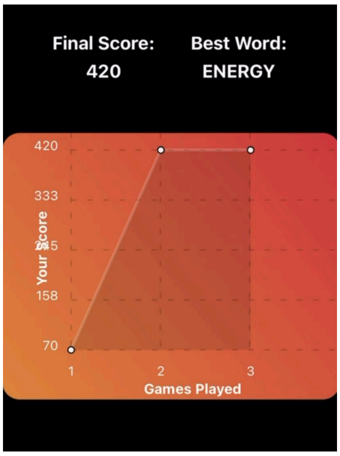


Figure 22 - Synchronous function for score saving

viewRef is set up as a reference to the part of the screen that the user will capture while layoutReady is used to ensure that the layout is fully rendered. Before capturing the image, the function asks the user for permission to access the device's media library. If granted, *captureRef* captures the screenshot and saves it to the device's gallery, with error handling included for any issues.

Figure 23 shows the libraries and functions imported to handle this. captureRef is used to capture a screenshot of a specific portion of the UI. The expo-media-library module provides access to the device's media library, allowing the application to manage and interact with the user's photos and videos.

```
// for screenshotting
import { captureRef } from 'react-native-view-shot';
import * as MediaLibrary from 'expo-media-library';
```

Figure 23 - Screenshot functionality imports

4.3.3 Design Implementations

4.3.3.1 SettingsContext.js

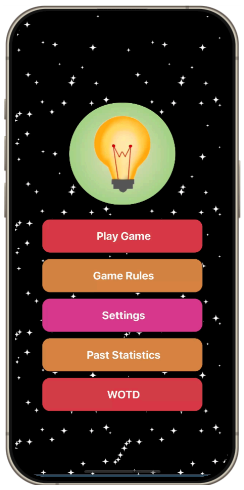
Application-wide settings like theme (light vs. dark mode), colour scheme (original vs. high contrast), and sound effects (on vs. off) are effectively managed by SettingsContext.js. These options are centrally located for easy access by pages, which streamlines the application's structure, guarantees a consistent user experience, and improves accessibility. It is also scalable for updates in the future.

```
export const SettingsContext = createContext();

export const SettingsProvider = ({ children }) => {
  const [darkMode, setDarkMode] = useState(true);
  const [highContrast, setHighContrast] = useState(false);
  const [soundEffects, setSoundEffects] = useState(true);
  const [notifications, setNotifications] = useState(true);
  return (
    <SettingsContext.Provider value={{ darkMode, setDarkMode,
                                       highContrast, setHighContrast,
                                       soundEffects, setSoundEffects,
                                       notifications, setNotifications }}>
      {children}
    </SettingsContext.Provider>
  );
};
```

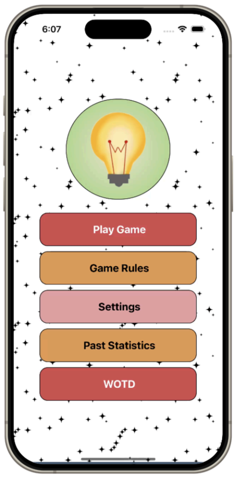
Figure 24 - Settings Context function

Figure 24 below shows the code in homescreen.js which controls colours of the different elements on the page. It also shows how the UI of the homepage is affected when the highContrast and darkMode settings are on or off.

```
<TouchableOpacity
style={[styles.navButtons, { backgroundColor: '#D24245' }]}
onPress={() => navigation.navigate('PlayGame')}>
  <Text style={styles.buttonText}> Play Game </Text>
</TouchableOpacity>

<TouchableOpacity
style={[styles.navButtons, { backgroundColor: highContrast ? '#e19046' : '#D28642' }]}
onPress={() => navigation.navigate('GameRules')}>
  <Text style={[styles.buttonText, { color: highContrast ? 'black' : 'white'}]}> Game Rules </Text>
</TouchableOpacity>

<TouchableOpacity
style={[styles.navButtons, { backgroundColor: highContrast ? '#e59698' : '#D2428D' }]}
onPress={() => navigation.navigate('Settings')}>
  <Text style={[styles.buttonText, { color: highContrast ? 'black' : 'white'}]}> Settings </Text>
</TouchableOpacity>

<TouchableOpacity
style={[styles.navButtons, { backgroundColor: highContrast ? '#e19046' : '#D28642' }]}
onPress={() => navigation.navigate('PastStatistics')}>
  <Text style={[styles.buttonText, { color: highContrast ? 'black' : 'white'}]}> Past Statistics </Text>
</TouchableOpacity>

<TouchableOpacity
style={[styles.navButtons, { backgroundColor: '#D24245' }]}
onPress={() => navigation.navigate('WOTD')}>
  <Text style={styles.buttonText}> WOTD </Text>
</TouchableOpacity>
```



highContrast: off
darkMode: on



highContrast: on
darkMode: off

Figure 25 - Contrast Code and Effects

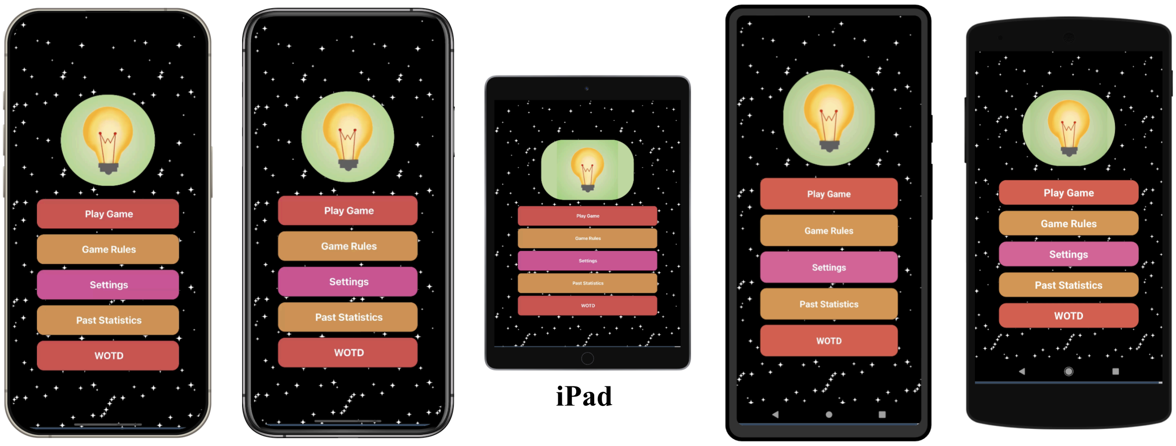
4.3.3.2 Responsive UI Elements

The goal of including responsive elements in the UI of WOrdEureka! Was to ensure that they can adapt to various screen sizes regardless of the device the game is played on in order to offer users a consistent and reliable user experience. wordEureka! adapts responsive design by designing all TouchableOpacity elements to scale in size and spacing depending on the screen dimensions. Flexbox properties such as flexDirection, justifyContent, and alignItems are also used to ensure that the layout of the elements are aligned and properly spaced, making them adaptable yet uniform across different screen sizes.

The benefits of using responsive UI elements are threefold:

- they allow for a consistent user experience no matter the device
- they ensure the application is easy to use regardless of screen size
- it contributes to an accessible user experience by ensuring users with different needs can comfortably use the app on any device

Figure 26 below shows the homepage of wordEureka! across devices of different screen sizes and operating softwares. It shows the consistency of the pages across the devices and portrays how responsive UI elements come into play in such scenarios.



iPhone 15 Pro **iPhone 11 Pro** **iPad** **Pixel 7** **Nexus 5**

Figure 26 - Various Phone Scaling Examples

Repository Link: https://snack.expo.dev/@shambhavi912/newwordeureka

Chapter 4 Word Count: 2069/2000

CHAPTER 5: EVALUATION

This chapter of the report provides a comprehensive evaluation of WordEureka!. It discusses the testing and evaluation methodologies employed to ensure the app's quality and functionality, as well as the results of user studies conducted to assess its impact on users. Examining the evaluation results will help provide a critical assessment of WordEureka!'s successes, limitations, and areas for future improvement.

Testing is a crucial phase in the development lifecycle, ensuring that the application functions as intended and meets the desired quality standards. This section covers three main types of testing conducted on WordEureka!: unit testing, integration testing, and user testing.

Following testing, the evaluation phase provides a critical analysis of the application based on user feedback and quantitative metrics. This section discusses the results obtained using the Perceived Stress Scale (PSS) and a customised User Feedback Scale. These offer valuable perspectives on user satisfaction, engagement, and areas for improvement, and on whether WordEureka! Has met the aims it set out to.

5.1 Testing

5.1.1 Unit & Integration Testing (white box testing)

The unit and integration tests were conducted through manual testing. The primary objective of the testing was to ensure that the mobile application performs as intended across various devices and operating systems. This was crucial in verifying its stability, functionality, and user experience and ensuring its compatibility across different screen sizes and software environments. A combination of 57 unit, functional, integration, and UI tests were conducted by myself and some of my fellow computer science classmates who had volunteered to assist me with the tests. These tests were conducted across four devices - iPhone 1 Pro, iPhone 8, Pixel 4, and Galaxy Note 24 Ultra. The detailed results of these tests are attached in Appendix E1.

A structured approach was taken where each test was executed manually based on predefined test cases. These cases covered critical aspects of the application, including core functionalities and device-specific features such as screen responsiveness and performance. A combination of the 4 above-mentioned types of tests allowed me to test individual functions and methods, that the different modules of the application interacted correctly, and that the UI rendered consistently across all the devices.

Out of the 57 tests conducted, the majority passed successfully, demonstrating the consistency and success of the application across multiple devices. However, some performance discrepancies were observed between older devices, such as the iPhone 8, and newer ones like theGalaxy Note 24 Ultra. This allowed for the necessary changes to be made accordingly in order to ensure that all the tests pass. Once these changes were made, a second round of manual testing was conducted across the same devices and the detailed results of the second round of testing are documented below in Appendix E1. In the second round of testing, all test cases passed across all devices.

The manual testing process was instrumental in ensuring that the mobile application functioned properly across different devices and operating systems. This thorough testing process has validated the app's stability, ensuring it delivers a reliable user experience regardless of device.

5.1.3 User Testing (black box testing)

After completing white-box testing, black-box testing was conducted during the final stages of development. This form of testing was essential to evaluate the functionality and user experience without focusing on the internal code structure. User testing was carried out using the System Usability Scale (SUS) [18] to collect feedback from potential users on the usability of WordEureka!. The list of questions used for this test is provided in the Appendix E2 [18].

Once responses were gathered, the SUS score was calculated following the guidelines outlined in [18]. These scores were plotted in a graph (Figure 27) to provide a visual representation of user feedback. According to [18], a SUS score of 80.3 or higher indicates users love the application and are likely to recommend it to others, while a score around 68 suggests the app is acceptable but could use some improvements. A score of 51 or lower signifies significant usability issues that need immediate attention.

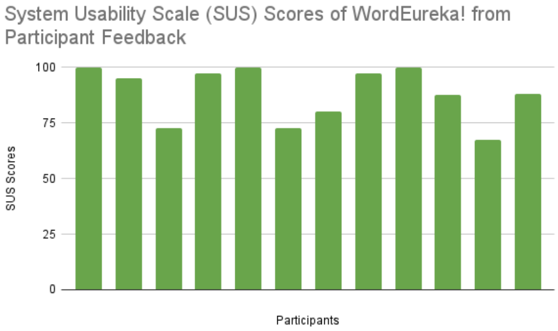


Figure 27 - graph showing different participants' SUS scores for WordEureka!

The average usability score for WordEureka! is 88.1 (Figure 28), with three users giving it a perfect score of 100. This indicates that the application is highly usable and easy to navigate. Additionally, as seen in Figure 27, the majority of participants rated WordEureka! with a SUS score of 80 or above, demonstrating consistent positive feedback across different users.

USER	FINAL SUS SCORE
User 1	100
User 2	95
User 3	72.5
User 4	97.5
User 5	100
User 6	72.5
User 7	80
User 8	97.5
User 9	100
User 10	87.5
User 11	67.5
User 12	
User 13	
User 14	
User 15	
Average	88.18181818

Figure 28 - SUS score summary table

5.2 Evaluation

5.2.1 Perceived Stress Scale

The Perceived Stress Scale (PSS) was used to measure the stress levels of users before and after playing WordEureka!. Initially, participants completed the standard PSS (see Appendix E2), which assessed their daily stress levels. The results, illustrated in Figure 29, show that 64.3% of users experienced moderate stress and 28.6% reported high stress, meaning that 92.9% of respondents were dealing with some level of stress in their daily lives.

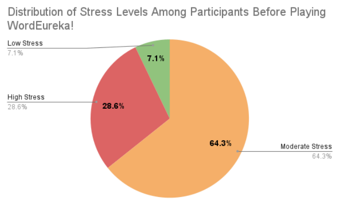


Figure 29 - stress distribution before playing game

After gameplay, participants completed a modified version of the PSS (Appendix E3) to gauge their stress levels post-gameplay. The purpose of this comparison was to determine if playing WordEureka! had any effect on reducing stress. The results, shown in Figure 30, indicate a significant improvement: 66.7% of users reported low stress levels (up from 7.1% before playing), and none of the participants reported high stress while playing the game.

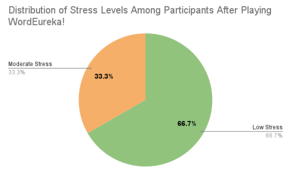


Figure 30 - stress distribution after playing game

To provide a more detailed comparison, Table 7 and Table 8 below highlight the differences in PSS scores before and after gameplay. The average PSS score before playing the game was 22.13, which decreased to 11.09 after gameplay, showcasing a notable reduction in stress levels.

	Overall Average PSS Scores
Before	22.13333333
After	11.09090909

Table 7 - comparison table showing average PSS scores of users before and after

	Before (%)	After (%)
Low Stress	7.1	63.6
Moderate Stress	64.3	36.4
High Stress	28.6	0

Table 8 - table showing 5 difference of stress levels of users before and after the game

These results clearly demonstrate that WordEureka! significantly reduces stress levels in users. A majority of users experienced a shift from moderate or high stress before playing the game to low stress after gameplay. The findings from the PSS tests indicate that WordEureka! has a positive impact on reducing stress. Prior to playing the game, nearly all users experienced moderate to high stress. After gameplay, there was a substantial reduction in stress levels, with the majority reporting low stress and none reporting high stress. This suggests that WordEureka! could be an effective tool for stress relief and may contribute positively to users' mental well-being.

5.2.2 User Feedback Survey

A user feedback form was also crafted in order to get more nuanced feedback from potential users. Refer to appendix E4 for the whole set of questions used in the survey. These questions were derived to evaluate different facets of WordEureka!'s success such as user satisfaction, cognitive engagement, feature utility, and its potential to act as an alternative to social media. Collecting data on these aspects is essential to determine whether the app fulfils its intended goals and to gauge its effectiveness in fulfilling the core aim of the project.

Three questions included in the survey address the issue of users' perception of WordEureka! in relation to social media usage and its potential as a tool for mental well-being. These questions are as follows:

1. How likely are you to play WordEureka! instead of scrolling on social media?
2. How far do you agree that WordEureka! can act as an accessible and useful alternative to excessive social media usage?
3. Do you think that WordEureka! can help you improve your mental health and combat feelings of brain rot?

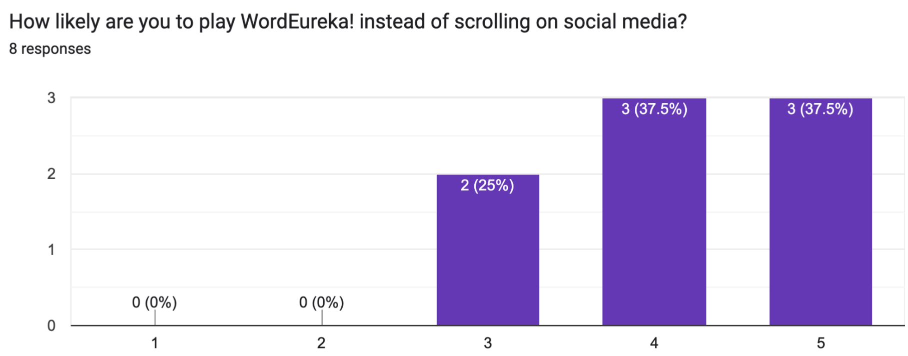


Figure 31 - likelihood of users choosing WordEureka! over social media usage

Figure 31 above shows that 75% of respondents agreed that they would be likely to choose WordEureka! over social media. This indicates that the game is serving its intended purpose as an appealing and cognitively stimulating alternative to social media, which can often contribute to mindless consumption and "brain rot."

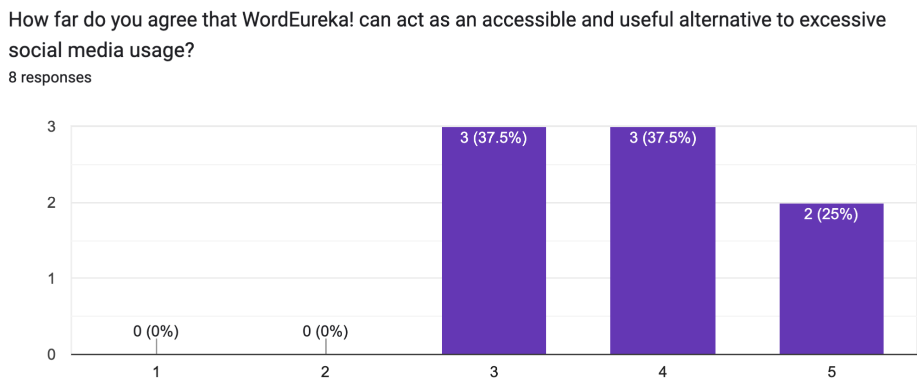


Figure 32 - user perception on WordEureka! being an alternative to social media usage

The question, "How far do you agree that WordEureka! can act as an accessible and useful alternative to excessive social media usage?", received 62.5% agreement from users. This feedback supports the notion that WordEureka! is accessible to its target demographic and offers a beneficial alternative to spending excessive time on social platforms. By offering a gamified learning experience, users feel more engaged, productive, and mentally stimulated, aligning with the app's goal of reducing unproductive screen time.

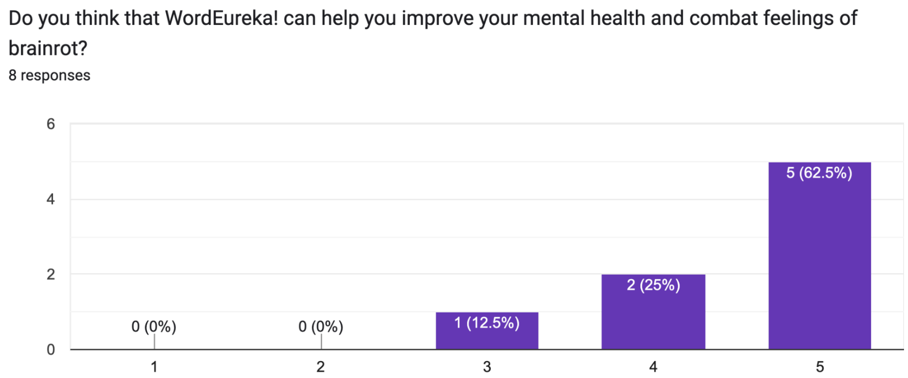


Figure 33 - users' perceived impact of WordEureka! on mental health and managing brain rot

Figure 33 above shows that a significant 87.5% of respondents agreed with the statement, "Do you think that WordEureka! can help you improve your mental health and combat feelings of brain rot?" This overwhelming positive response demonstrates that users believe the game has real potential to improve their mental well-being. This is a key indicator that the project is succeeding in its mission to promote cognitive stimulation and alleviate the mental fatigue associated with excessive social media use.

5.3 Summary

This chapter evaluated WordEureka! through technical testing and user feedback. Unit and integration testing across multiple devices confirmed the app's functionality and stability, with all test cases passing successfully after two rounds of manual testing. The System Usability Scale (SUS) gave WordEureka! a high average score of 88.1, well above the industry benchmark of 68. This shows that users found WordEureka! highly usable and easy to navigate and indicates strong user satisfaction.

The Perceived Stress Scale (PSS) results showed that playing WordEureka! significantly reduced users' stress levels, with the average score dropping from 22.13 to 11.09, indicating that the game had a meaningful impact on alleviating users' stress.

Additionally, user feedback also strongly supported the main aim of the project as 75% of users stated they would play WordEureka! over scrolling on social media, 62.5% believed it could serve as an accessible and useful alternative to social media, and 87.5% agreed that the game could improve their mental health and combat "brain rot."

Overall, the app effectively meets its goals of offering an engaging alternative to social media and improving mental well-being, thus validating the success of WordEureka! in achieving its core objectives. This shows that the app offers a valuable, engaging, and mentally stimulating experience for users, with tangible benefits in stress reduction and cognitive engagement. These findings indicate that WordEureka! has the potential to positively impact its target demographic, particularly in helping to reduce excessive social media use and improve mental well-being.

Chapter 5 Word Count: 1623/1500

CHAPTER 6: CONCLUSION

6.1 Summary

WordEureka! was ideated to tackle the issue of excessive social media usage among Singaporean youths by aiming to be an accessible and useful alternative to such platforms. Every design and functional aspect of the application was aimed to address the prevalent issue of mental fatigue induced by prolonged, unproductive screen time, ie brain rot. The iterative development process involved testing and user feedback to ensure the application was both functional and user-friendly. The success of wordeureka was evaluated through well known metrics such as the System Usability Scale (SUS) and Perceived Stress Scale (PSS),

6.2 Outcome (evaluation & testing)

The evaluation and testing phase showed that WordEureka! successfully met its objectives. A variety of tests were conducted through manual testing to ensure the stability and functionality of the app across multiple devices, with all test cases passing after iterative testing. WordEureka also scored a high 88.1 on the SUS scale, reflecting a high level of usability and user satisfaction. Additionally the PSS scores decreasing from 22.13 (pre-game) to 11.09 (post-game) also confirm that the app is technically sound and effectively serves its purpose in enhancing users' mental well-being.

6.3 Impact on Target Audience

The user feedback collected shows that WordEreka was a hit amongst the potential users as 75% of users mentioned they would choose WordEureka! over social media as a mental health-enhancing alternative, while 62.5% agreed that the app could potentially replace excessive social media usage. Additionally, 87.5% of participants believed that the game had a positive impact on their mental health, highlighting the app's potential to help users combat stress and 'brain rot.' These figures underscore WordEureka!'s success in delivering its core mission of providing a stimulating, stress-relieving experience for its target demographic of Singaporean youths.

6.4 Challenges & Limitations

Despite its success among users, the app initially faced certain limitations when played on older devices as they needed additional adjustments. Additionally, the scope of user testing was quite limited as the tests were conducted using a small number of participants. Expanding this testing to a broader and more diverse audience could provide more comprehensive feedback. Additionally, while WordEureka! effectively reduced stress for most users, it may not address all aspects of mental well-being, and future iterations should explore ways to further enhance its cognitive engagement features.

6.5 Further Work

If the application were to grow in popularity, it would require more nuanced storage abilities. Therefore, a cloud storage option such as Firebase is highly recommended. Furthermore, in order to reach out to a bigger audience group, the application could possibly be made available in different languages so even non-English speakers can enjoy WordEureka!.

From a technical perspective, implementing more rigorous automated testing frameworks would further enhance its stability and compatibility across an even wider range of devices. Future work should also focus on conducting long-term studies to measure the sustained effects of WordEureka! on mental health and its role in reducing social media usage. Expanding the user base and gathering insights from a more diverse demographic would provide valuable data to refine the app further.

Chapter 6 Word Count: 522/1000

Overall Report Word Count: 9124/9500

CHAPTER 7: APPENDICES

Appendix A

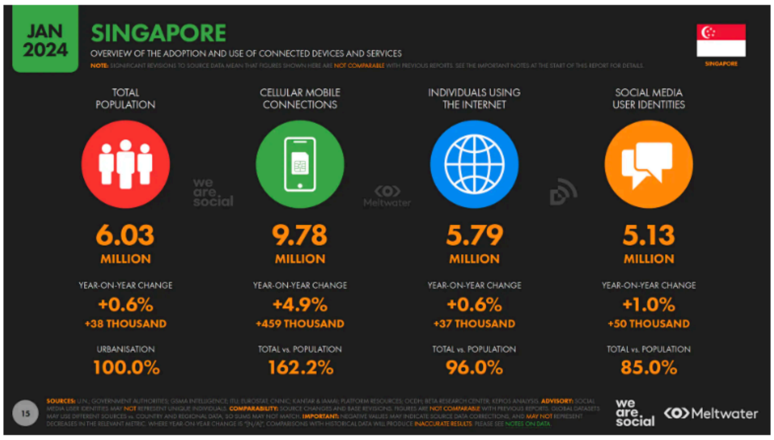


Figure A1 - Overview of the adoption and use of connected devices and services [Source: Sue Howe. 2024. *Meltwater* (May 2024) (Figure in source not titled)]

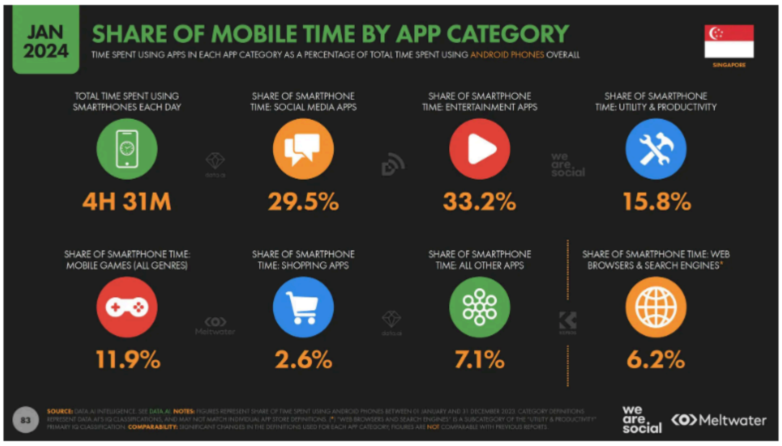


Figure A3 - Share of mobile time by App category [Source: Sue Howe. 2024. *Meltwater* (May 2024) (Figure in source not titled)]

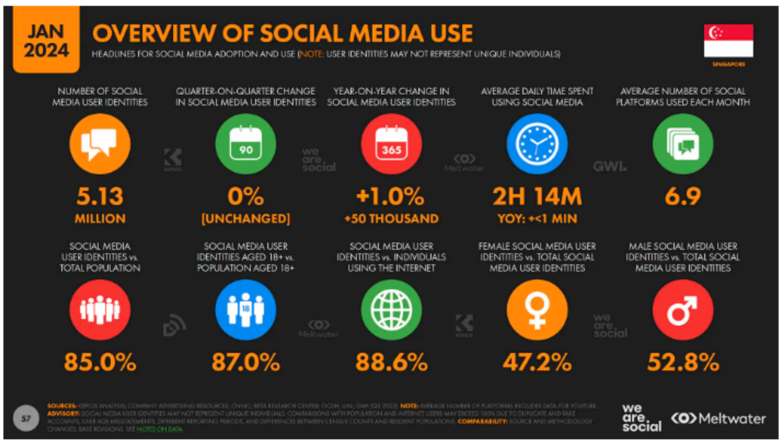


Figure A3 - Overview of Social Media use [Source: Sue Howe. 2024. *Meltwater* (May 2024) (Figure in source not titled)]

Appendix B

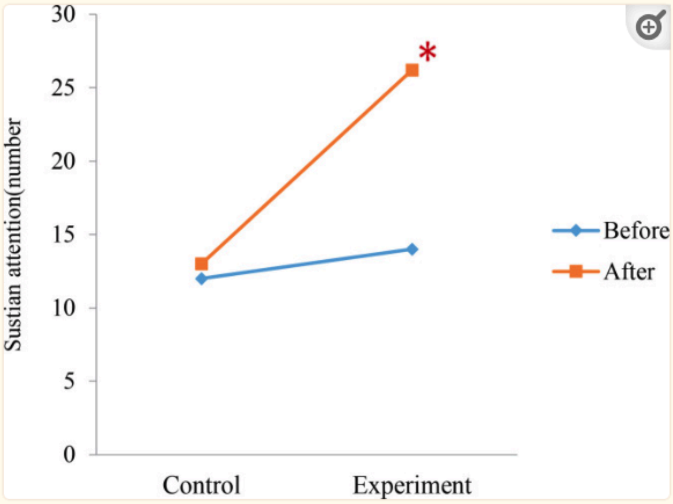


Figure 3.

Changes in sustained attention in the experimental and control groups Comparing sustained attention among participants before and after the intervention. *P≤0.05.

Figure B1 - Changes in sustained attention in the experimental and control groups, [Source: Hamed Aliyari, Hedayat Sahraei, Sahar Golabi, Masoomeh Kazemi, Mohammad Reza Daliri, and Behrouz Minaei-Bidgoli. 2021. The effect of brain teaser games on the attention of players based on hormonal and brain signals changes. (September 2021).]

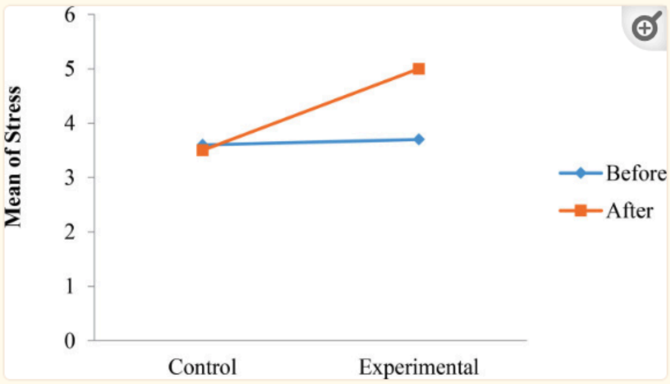


Figure 7.

Changes in stress level in the experimental and control groups

Mean changes in the power of stable brainwaves energy, construct of stress among players

Figure B2 - Changes in stress level in the experimental and control groups, [Source: Hamed Aliyari, Hedayat Sahraei, Sahar Golabi, Masoomeh Kazemi, Mohammad Reza Daliri, and Behrouz Minaei-Bidgoli. 2021. The effect of brain teaser games on the attention of players based on hormonal and brain signals changes. (September 2021).]

Appendix C

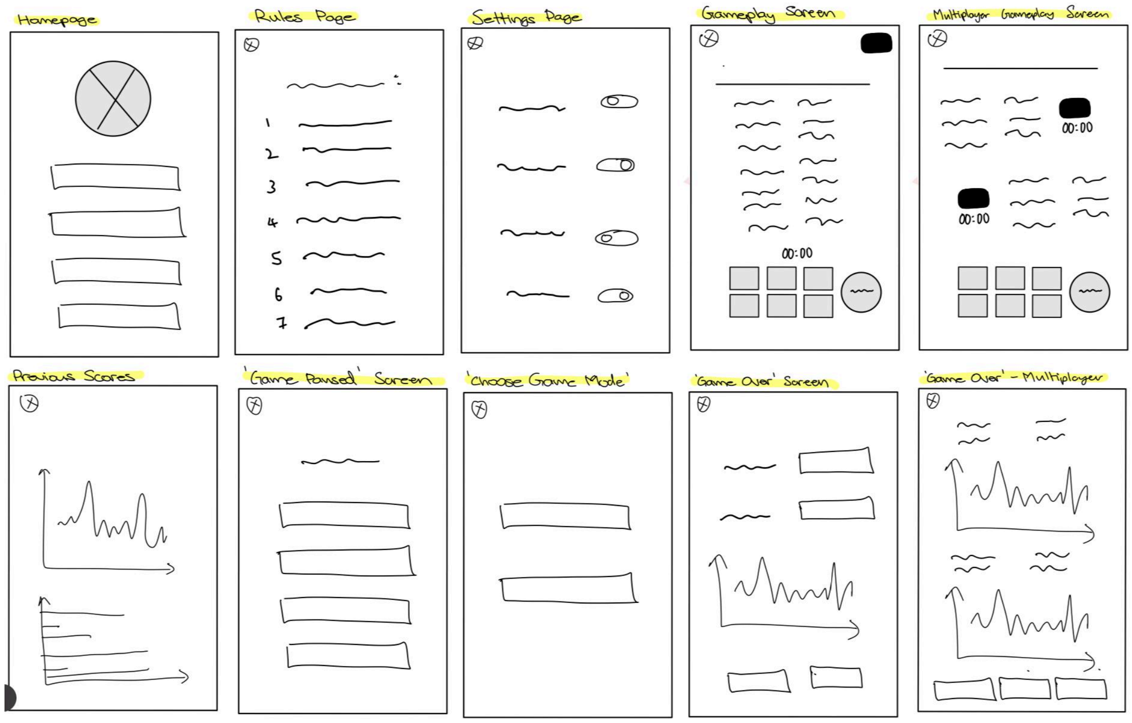


Figure C1 - Low fidelity wireframe of WordEureka!

Figure C2 - Mid fidelity wireframe of WordEureka!

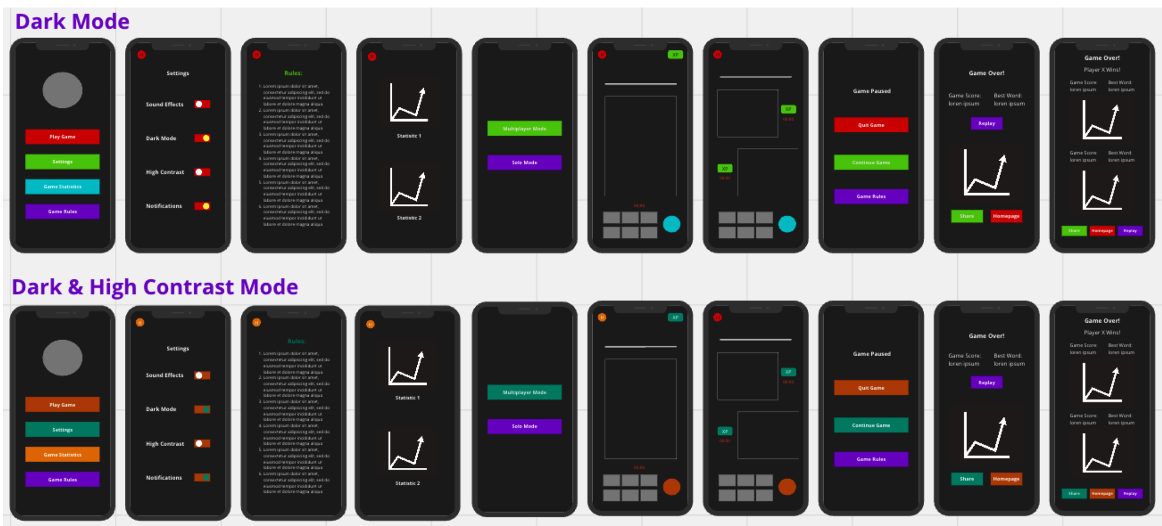


Figure C3 - High fidelity wireframe (dark & high contrast mode) of WordEureka!

Appendix D

How do you find the layout of the homepage? Any improvements to be made? (Colours, layout etc.)	How do you find the layout of the settings page? Any improvements to be made? (Colours, layout etc.)	How do you find the layout of the rules page? Any improvements to be made? (Colours, layout etc.)	How do you find the layout of the choose gameplay page? Any improvements to be made? (Colours, layout etc.)	How do you find the layout of the game paused page? Any improvements to be made? (Colours, layout etc.)	How do you find the layout of the game results page? Any improvements to be made? (Colours, layout etc.)
Simple but effective	looks good, perhaps tweak the colour hues	Looks good	Looks good, perhaps some designs in the background or theme could be nice in the future	Perhaps add a settings button?	Interesting
seems ait simple maybe can use rounded corners	too much space in between	too many words	maybe can be a small window instead of a whole page	can be translucent to still show the paused gameplay	looks ok
background looks very plain. make it more attractive by including more designs and colours	settings page seems fine altho i wud add more colour to it	rules page seems fine. this can be simple so no worries	add more diagrams	this one is okay	this is perfect
its nicee	maybe instead of usign the rectangles can use the rounded button slider instead	simple, straight to the point	nice! dont really have any suggestions for improvements	simple and effective	nice and clear, no improvements
colourful. chamfer the buttons to be round might be more appealing	yellow and red contrast, not so nice. perhaps can use ocean blue to replace the red background. layout wise, each row dont spread out so wide, make is narrower then place it in the center would be better.	would be better if could include some images after every rule pointer, might be good for those who are more of a visual learner	neon green, might be too bright. slightly darker so it will look nicer. otherwise can just choose one colour but one is lighter than the other.	this is ok	this is ok too
colours are striking and buttons are clear	maybe the circle can be green indicating it is on	simple to understand and straight to the point	clear to see and understand	none that i can think of	nope i think it's pretty
Very boring. Colour theme can be changes according to the game. Font is too simple and common.	Same changes as home page	It's like a paragraph. Most will skip it's like this. Make it concise.	Only colour change needed.	Same colour needed to be changed according to game theme.	Many people won't understand graphics. Can be changed to score system.
Pretty solid, maybe a consistent colour scheme cld be adopted to give the game a certain look and feel	Layout is good but somewhat sparse	👍	White against green for the top button is not contrasting enough	Na	Na
The layout of the homepage is fantastic It's clean, organized, and makes it really easy to find what I'm looking for.	the settings page is really intuitive and user-friendly	It's well-organized and makes the information easy to read and understand	It makes selecting different gameplay options straightforward and enjoyable	its excellent	it easy to quickly understand the outcome and see the key statistics
Layout is clear maybe could be made a bit more aesthetic? Rounded corners for buttons or smth?	Could make the rectangles green when on so it's clearer? Yellow and white may be hard to distinguish	Looks good	Looks good, maybe have a title for the page - like choose your mode or smth?	Looks good	Looks good

How do you find the layout of the gameplay page? Any improvements to be made? (Colours, layout etc.)	Do you have any feedback or suggestions for the overall user interface of the game?	The above image shows all of the pages in the game in Dark Mode and High Contrast Mode. How do you find the layout of these pages? Any improvements to be made? (Colours, layout, cohesion etc.)	The above image shows all of the pages in the game in Light Mode and High Contrast Mode. How do you find the layout of these pages? Any improvements to be made? (Colours, layout, cohesion etc.)	Do you have any other feedback or suggestions for the different overall user interfaces of the game?
Nice	None	I like it	Looks good	none
the red is a bit too intense against the black	overall looks fine but maybe can move away from dark theme bcz the colours might look abit too serious	looks ok	maybe can desaturate the colour?	dont think so
looks very plain. black doesnt look nice. maybe add more colours	game seems fine. just make it look more attractive	black just seems olain. maybe add more colour to the black (like designs)	Light mode looks better.	na
nice!	i'm sure it will look nice when its made! look forward to playtesting it :)	high contrast mode is good, colours are not glaring, good for night usage	all's good, would definitely use the high contrasts mode personally since its less glaring. maybe for the light mode settings page shouldnt use neon blue for the slider	nope!
this is ok too	if got time, maybe can have one in light mode	overall looks ok, the rest i have mentioned in the previous	this is ok too, the colour scheme seems to fit nicer on the light mode rather than the dark.	i would say the colour scheme plays an important role cos now as a whole it looks pretty individual but then it is very subjective as well. i think one thing is to look back at the domain users and from there see which age gap are you targeting at. hence, with that in mind then you can try to adjust the colours that will appeal to them.
i think it captures my attention well	no i think it is done effectively and in a simple manner. it is user friendly	the colours used are standard and all fits the theme. they are able to be seen clearly on a dark background.	the colours used are standard and all fits the theme. they are able to be seen clearly on a light background.	no i think the colours are a smart choice such that they work well on both light and dark background.
Design is too old. Layout can be changed	This format is very simple and boring. Need more improvement to attract gamers.	Use only two or three colours max. Layout is too simple	Same as above. It look bland.	Make it more interesting
Na	Na	Could be made more cohesive with a fixed colour scheme and elements to beautify the screen	Same as above	Na
The page is already quite effective	The overall user interface of the game is impressive. It's visually appealing and user-friendly	i think its nice to have a dark mode and its impressive	i prefer the dark mode	nope. its amazing
Looks good	The layout is very clear and intuitive	Looks good	Looks good	nope

Figure D1 -

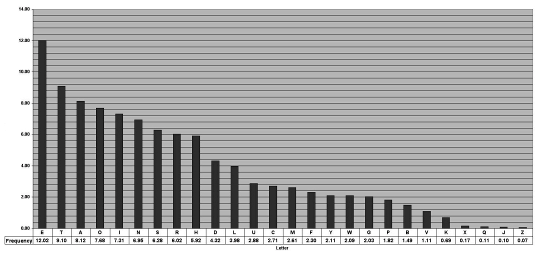
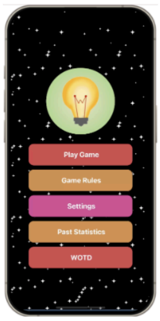
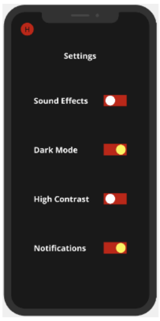
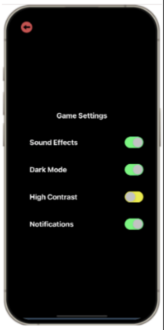
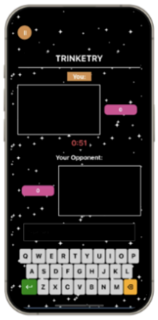
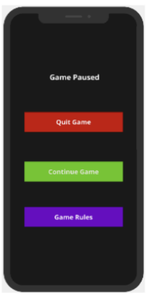


Figure D2

Homepage		
Settings Page		

Gameplay page		
		

Rules Page		
Choose Gameplay page		
Game Paused Page		

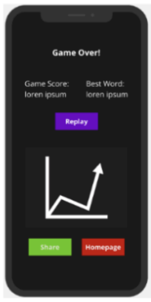
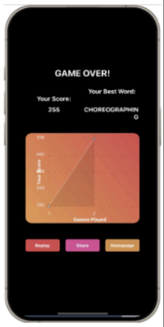
Results Page		
		
Gameplay page		

Table D1

Appendix E

Test Case ID	Description	Type of test	Steps	Expected Outcome	Status			
					iPhone 15 Pro	iPhone 8	Pixel 4	Galaxy Note 24 Ultra
HOMEPAGE								
A1	Verify that the intro sound pla	functional	Open the app on the device. Listen for the intro sound.	The intro sound should play when the app is opened.	Pass	Pass	Pass	Pass
A2	Verify that the UI displays correctly on the homepage.	UI	Open the app. Observe the layout and elements on the homepage.	The UI should be displayed correctly without any layout issues.	Pass	Pass	Pass	Pass
A3	Verify that the user can navigate to the play game screen and return to the homepage.	functional	From the homepage, navigate to the play game screen. Return to the homepage.	The user should be able to navigate to the play game screen and return to the homepage without issues.	Pass	Pass	Pass	Pass
A4	Verify that the user can navigate to the rules screen and return to the homepage.	functional	From the homepage, navigate to the rules screen. Return to the homepage.	The user should be able to navigate to the rules screen and return to the homepage without issues.	Pass	Pass	Pass	Pass
A5	Verify that the user can navigate to the settings screen and return to the homepage.	functional	From the homepage, navigate to the settings screen. Return to the homepage.	The user should be able to navigate to the settings screen and return to the homepage without issues.	Pass	Pass	Pass	Pass
A6	Verify that the user can navigate to the stats screen and return to the homepage.	functional	From the homepage, navigate to the stats screen. Return to the homepage.	The user should be able to navigate to the stats screen and return to the homepage without issues.	Pass	Pass	Pass	Pass
A7	Verify that the user can navigate to the word of the day (wotd) screen and return to the homepage.	functional	From the homepage, navigate to the word of the day (wotd) screen. Return to the homepage.	The user should be able to navigate to the wotd screen and return to the homepage without issues.	Pass	Pass	Pass	Pass
SOLO								
B1	Verify that the user can choose between solo or 1v1 modes and that the countdown begins.	functional	Launch the game. Choose between solo mode or 1v1 mode. Verify that the countdown starts.	The game should start a countdown when the user selects solo or 1v1 mode.	Pass	Pass	Pass	Pass
B2	Verify that navigating to solo game mode fetches a random word to begin word play.	functional	Navigate to the solo game mode. Verify that a random word is fetched for gameplay.	A random word should be fetched when entering solo game mode.	Pass	Pass	Pass	Pass
B3	Verify that the user can pause the game.	functional	Start the game. Pause the game.	The game should pause successfully.	Pass	Pass	Pass	Pass
B4	Verify that the user can resume the game and that stats are paused and resumed correctly.	functional, integration	Start and pause the game. Resume the game. Verify that game stats are correctly paused and resumed.	The game should resume correctly, and stats should reflect the paused state accurately.	Pass	Pass	Pass	Pass
B5	Verify that the user can restart the game and that stats are reset.	functional, integration	Start and play the game. Restart the game. Verify that the game stats are reset.	The game should restart, and all stats should be reset.	Pass	Pass	Pass	Pass
B6	Verify that the user can navigate to the rules screen from the pause menu and return to the game.	functional	Pause the game. Navigate to the rules screen. Return to the paused game.	The user should be able to view the rules and return to the paused game without issues.	Pass	Pass	Pass	Pass
B7	Verify that the user can quit the game and that stats are not saved upon quitting.	functional	Start the game. Quit the game. Verify that the game stats are not saved.	The game should quit, and no stats should be saved.	Pass	Pass	Pass	Pass
B8	Verify that the game ends and a popup appears when the timer hits 0.	functional	Play the game until the timer reaches 0. Verify that the game ends and a popup appears.	The game should end, and a popup should appear when the timer hits 0.	Pass	Pass	Pass	Pass
B9	Verify that the user can replay the game after it ends and that stats are reset.	functional	Play the game until it ends. Click replay. Verify that the game restarts and stats are reset.	The game should restart, and stats should be reset.	Pass	Pass	Pass	Pass
B10	Verify that the user can share their score after the game ends.	functional	Play the game until it ends. Share the score. Verify that the score is shared successfully.	The score should be shareable after the game ends.	Pass	Pass	Pass	Pass
B11	Verify that the user can return to the home screen after the game ends.	functional	Play the game until it ends. Click the button to go home. Verify that the user is taken to the home screen.	The user should be able to navigate to the home screen after the game ends.	Pass	Pass	Pass	Pass
B12	Verify that the best score and best word are displayed on the game end page.	integration	Play the game until it ends. Check the game end page. Verify that the best score and best word are displayed.	The best score and best word should be displayed on the game end page.	Pass	Pass	Pass	Pass
B13	Verify that the graph is updated on the game end page.	integration	Play the game until it ends. Check the game end page. Verify that the graph is updated with the latest game data.	The graph should be updated on the game end page.	Pass	Pass	Pass	Pass
B14	Verify that the user can submit a word, and the scores are updated correctly with feedback.	functional, unit	Enter a word during gameplay. Submit the word. Verify that the scores are updated correctly and feedback is provided.	The word should be accepted, scores should be updated correctly, and feedback should be provided.	Pass	Pass	Pass	Pass
B15	Verify that if the same word is entered again, it is not accepted, and no scores or feedback are given.	functional, unit	Enter a word during gameplay. Enter the same word again. Verify that the word is not accepted, no scores are updated, and no feedback is given.	The same word should not be accepted again, and there should be no score update or feedback.	Pass	Pass	Pass	Pass
B16	Verify that if an invalid word is entered, it is not accepted, and no scores or feedback are given.	functional, unit	Enter an invalid word during gameplay. Verify that the word is not accepted, no scores are updated, and feedback is given.	Invalid words should not be accepted, no scores should be updated, and feedback should be provided.	Pass	Pass	Pass	Pass
B17	Verify that if the letters in the entered word do not match the required letters, the word is not accepted, and no scores or feedback are given.	functional, unit	Enter a word with letters that do not match the required letters during gameplay. Verify that the word is not accepted, no scores are updated, and feedback is provided.	Words with incorrect letters should not be accepted, no scores should be updated, and feedback should be provided.	Pass	Pass	Pass	Pass
B18	Verify that the UI displays correctly during gameplay.	functional, UI	Start the game. Observe the UI elements during gameplay.	The UI should display correctly without any layout or functional issues.	Pass	Pass	Pass	Pass
1v1								

C1	Verify that the user can choose between solo or 1v1 modes and that the countdown begins.	functional	Launch the game. Choose between solo mode or 1v1 mode. Verify that the countdown starts.	The game should start a countdown when the user selects solo or 1v1 mode.	Pass	Pass	Pass	Pass	
C2	Verify that navigating to 1v1 game mode fetches a random word to begin word play.	functional	Navigate to the 1v1 game mode. Verify that a random word is fetched for gameplay.	A random word should be fetched when entering 1v1 game mode.	Pass	Pass	Pass	Pass	
C3	Verify that the user can pause the game in 1v1 mode.	functional	Start the game in 1v1 mode. Pause the game.	The game should pause successfully.	Pass	Pass	Pass	Pass	
C4	Verify that the user can resume the game and that stats are paused and resumed correctly in 1v1 mode.	functional, integration	Start and pause the game in 1v1 mode. Resume the game. Verify that game stats are correctly paused and resumed.	The game should resume correctly, and stats should reflect the paused state accurately.	Pass	Pass	Pass	Pass	
C5	Verify that the user can restart the game and that stats are reset in 1v1 mode.	functional, integration	Start and play the game in 1v1 mode. Restart the game. Verify that the game restarts and stats are reset.	The game should restart, and all stats should be reset.	Pass	Pass	Pass	Pass	
C6	Verify that the user can navigate to the rules screen from the pause menu and return to the game in 1v1 mode.	functional	Pause the game in 1v1 mode. Navigate to the rules screen. Return to the paused game.	The user should be able to view the rules and return to the paused game without issues.	Pass	Pass	Pass	Pass	
C7	Verify that the user can quit the game and that stats are not saved upon quitting in 1v1 mode.	functional	Start the game in 1v1 mode. Quit the game. Verify that the game stats are not saved.	The game should quit, and no stats should be saved.	Pass	Pass	Pass	Pass	
C8	Verify that the game ends and a popup appears when the timer hits 0 in 1v1 mode.	functional	Play the game in 1v1 mode until the timer reaches 0. Verify that the game ends and a popup appears.	The game should end, and a popup should appear when the timer hits 0.	Pass	Pass	Pass	Pass	
C9	Verify that the user can replay the game after it ends and that stats are reset in 1v1 mode.	functional	Play the game in 1v1 mode until it ends. Click replay. Verify that the game restarts and stats are reset.	The game should restart, and stats should be reset.	Pass	Pass	Pass	Pass	
C10	Verify that the user can share their score after the game ends in 1v1 mode.	functional	Play the game in 1v1 mode until it ends. Share the score. Verify that the score is shared successfully.	The score should be shareable after the game ends.	Pass	Pass	Pass	Pass	
C11	Verify that the user can return to the home screen after the game ends in 1v1 mode.	functional	Play the game in 1v1 mode until it ends. Click the button to go home. Verify that the user is taken to the home screen.	The user should be able to navigate to the home screen after the game ends.	Pass	Pass	Pass	Pass	
C12	Verify that the best score and best game are displayed on the game end page in 1v1 mode.	integration	Play the game in 1v1 mode until it ends. Check the game end page. Verify that the best score and best game are displayed.	The best score and best game should be displayed on the game end page.	Pass	Pass	Pass	Pass	
C13	Verify that the graph is updated on the game end page in 1v1 mode.	integration	Play the game in 1v1 mode until it ends. Check the game end page. Verify that the graph is updated with the latest game data.	The graph should be updated on the game end page	Pass	Pass	Pass	Pass	
C14	Verify that the user can submit a word, and the scores are updated correctly with feedback in 1v1 mode.	functional, unit	Enter a word during gameplay in 1v1 mode. Submit the word. Verify that the scores are updated correctly and feedback is provided.	The word should be accepted, scores should be updated correctly, and feedback should be provided.	Pass	Pass	Pass	Pass	
C15	Verify that if the same word is entered again in 1v1 mode, it is not accepted, and no scores or feedback are given.	functional, unit	Enter a word during gameplay in 1v1 mode. Enter the same word again. Verify that the word is not accepted again, no scores are updated, and no feedback is given.	The same word should not be accepted again, and there should be no score update or feedback.	Pass	Pass	Pass	Pass	
C16	Verify that if an invalid word is entered in 1v1 mode, it is not accepted, and no scores or feedback are given.	functional, unit	Enter an invalid word during gameplay in 1v1 mode. Verify that the word is not accepted, no scores are updated, and feedback is provided.	Invalid words should not be accepted, no scores should be updated, and feedback should be provided.	Pass	Pass	Pass	Pass	
C17	Verify that if the letters in the entered word do not match the required letters, the word is not accepted, and no scores or feedback are given in 1v1 mode.	functional, unit	Enter a word with letters that do not match the required letters during gameplay in 1v1 mode. Verify that the word is not accepted, no scores are updated, and feedback is provided.	Words with incorrect letters should not be accepted, no scores should be updated, and feedback should be provided.	Pass	Pass	Pass	Pass	
C18	Verify that the UI displays correctly during gameplay in 1v1 mode.	UI test	Start the game in 1v1 mode. Observe the UI elements during gameplay.	The UI should display correctly without any layout or functional issues.	Pass	Pass	Pass	Pass	
C19	Verify that the 1v1 game mode changes accordingly when a word is not accepted and that the player is highlighted correctly.	functional, integration	Play the game in 1v1 mode. Enter a word that is not accepted. Verify that the player who should be highlighted is highlighted correctly.	The game mode should update correctly, and the current player's turn should be highlighted.	Pass	Pass	Pass	Pass	
				GAME RULES					
D1	Verify that the user can see the game rules.	functional	Navigate to the game rules section from the main menu or game settings. Check if the game rules are displayed correctly.	The game rules should be visible and correctly displayed.	Pass	Pass	Pass	Pass	
D2	ui displays correctly	unit	Verify that the UI displays correctly in the game rules section.	Navigate to the game rules section. Observe the UI elements for layout and design accuracy.	Pass	Pass	Pass	Pass	
				SETTINGS					
E1	Verify that toggling between light and dark mode updates the UI accordingly on every page.	functional	Toggle between light and dark mode in the settings. Navigate through various pages of the app. Check if the UI updates correctly according to the selected mode.	The UI should update to reflect the selected light or dark mode on every page.	Pass	Pass	Pass	Pass	
E2	Verify that toggling between high contrast modes updates the UI accordingly on every page.	functional	Toggle between high contrast modes in the settings. Navigate through various pages of the app. Check if the UI updates correctly according to the selected high contrast mode.	The UI should update to reflect the selected high contrast mode on every page.	Pass	Pass	Pass	Pass	

E3	Verify that toggling between sound settings updates the audio settings accordingly on every page.	functional	Toggle between sound settings (e.g., on/off) in the settings. Navigate through various pages and interact with sound-related features. Check if the sound settings are applied consistently across the app.	The sound settings should update and be applied correctly across the app.	Pass	Pass	Pass	Pass	
E4	Verify that the UI displays correctly in the settings section.	unit	Navigate to the settings section. Observe the UI elements for layout and design accuracy.	The UI should be correctly displayed without any layout or design issues.	Pass	Pass	Pass	Pass	
PAST STATISTICS									
F1	Verify that multiplayer statistics update correctly every time.	functional	Play multiple multiplayer games. Check if the multiplayer statistics are updated correctly after each game. Verify if the statistics display who wins and other relevant details.	Multiplayer statistics should update correctly and display accurate information about the game results.	Pass	Pass	Pass	Pass	
F2	Verify that solo game statistics update correctly every time.	functional	Play multiple solo games. Check if the solo game statistics are updated correctly after each game.	Solo game statistics should update correctly after each game.	Pass	Pass	Pass	Pass	
F3	Verify that multiplayer statistics display "nothing" when there are no statistics.	functional	Ensure no multiplayer games have been played. Check the multiplayer statistics section. Verify if it correctly displays "nothing" or an appropriate message indicating the absence of statistics.	The multiplayer statistics section should display "nothing" or an appropriate message when there are no statistics.	Pass	Pass	Pass	Pass	
F4	Verify that solo game statistics display "nothing" when there are no statistics.	functional	Ensure no solo games have been played. Check the solo game statistics section. Verify if it correctly displays "nothing" or an appropriate message indicating the absence of statistics.	The solo game statistics section should display "nothing" or an appropriate message when there are no statistics.	Pass	Pass	Pass	Pass	
F5	Verify that the UI displays correctly in the past statistics section.	unit	Navigate to the past statistics section. Observe the UI elements for layout and design accuracy.	The UI should be correctly displayed without any layout or design issues.	Pass	Pass	Pass	Pass	
WOTD									
G1	Verify that navigating to the WOTD page fetches a word from the API and displays the information correctly.	functional	Navigate to the Word of the Day (WOTD) page. Check if a word is fetched from the API. Verify that the word and related information are displayed correctly.	A word should be fetched from the API and displayed correctly on the WOTD page.	Pass	Pass	Pass	Pass	
G2	Verify that the UI displays correctly on the WOTD page.	unit	Navigate to the Word of the Day (WOTD) page. Observe the UI elements for layout and design accuracy.	The UI should be correctly displayed without any layout or design issues.	Pass	Pass	Pass	Pass	

System Usability Scale (SUS)						
S/N	Questions	Strongly Disagree				Strongly Agree
		1	2	3	4	5
1	I think that I would like to use this system frequently.					
2	I found the system unnecessarily complex.					
3	I thought the system was easy to use.					
4	I think that I would need the support of a technical person to be able to use this system.					
5	I found the various functions in this system were well integrated.					
6	I thought there was too much inconsistency in this system.					
7	I would imagine that most people would learn to use this system very quickly.					
8	I found the system very cumbersome to use.					
9	I felt very confident using the system.					
10	I needed to learn a lot of things before I could get going with this system.					

Figure E2

Pre-Game Perceived Stress Scale (PSS)						
S/N	Questions	Never				Very Often
		0	1	2	3	4
1	In the last month, how often have you been upset because of something that happened unexpectedly?					
2	In the last month, how often have you felt that you were unable to control the important things in your life?					
3	In the last month, how often have you felt nervous and stressed?					
4	In the last month, how often have you felt confident about your ability to handle your personal problems?					
5	In the last month, how often have you felt that things were going your way?					
6	In the last month, how often have you found that you could not cope with all the things that you had to do?					
7	In the last month, how often have you been able to control irritations in your life?					
8	In the last month, how often have you felt that you were on top of things?					
9	In the last month, how often have you been angered because of things that happened that were outside of your control?					
10	While playing the game, how often have you felt that the game's difficulties were piling up to a point where you couldn't overcome them?					

Figure E3

Post Game Perceived Stress Scale (PSS)						
S/N	Questions	Never				Very Often
		0	1	2	3	4
1	While playing the game, how often have you been upset because of something that happened unexpectedly in the game?					
2	While playing the game, how often have you felt that you were unable to control the important aspects of the gameplay?					
3	While playing the game, how often have you felt nervous or stressed by the challenges presented?					
4	While playing the game, how often have you felt confident about your ability to handle the game's challenges?					
5	While playing the game, how often have you felt that the gameplay was going your way?					
6	While playing the game, how often have you found that you could not cope with the tasks or challenges within the game?					
7	While playing the game, how often have you been able to control your frustrations or irritations?					
8	While playing the game, how often have you felt that you were on top of the gameplay and challenges?					
9	While playing the game, how often have you been angered because of things that happened in the game that were outside of your control?					
10	While playing the game, how often have you felt that the game's difficulties were piling up to a point where you couldn't overcome them?					

Figure E4

User Feedback						
S/N	Questions	Strongly Disagree				Strongly Agree
		1	2	3	4	5
1	Did you enjoy playing WordEureka!?					
2	Did you feel cognitively stimulated whilst paying the game?					
3	Would you recommend WordEureka! to friends/family?					
4	Did you find the customisable settings useful?					
5	How likely are you to play WordEureka! on a regular basis?					
6	How likely are you to play WordEureka! instead of scrolling on social media?					
7	How far do you agree that WordEureka! can act as an accessible and useful alternative to excessive social media usage?					
8	Do you think that WordEureka! can help you improve your mental health and combat feelings of brainrot?					
		Very Difficult				Very Easy
		1	2	3	4	5
9	How difficult did you find the game?					
		Very Useless				Very Useful
		1	2	3	4	5
10	How useful did you find the different features of the game?					

Figure E5

CHAPTER 8: REFERENCES

1. Adobe. 2022. User flow diagram — what it is, why it's important, and how to create one. (November 2022). Retrieved June 16, 2024 from https://business.adobe.com/blog/basics/how-to-make-a-user-flow-diagram
2. Anon. 2011. Spelltower Review. (December 2011). Retrieved May 28, 2024 from https://web.archive.org/web/20120104155447/http://www.next-gen.biz/reviews/spelltower-review
3. Anon. Gamer reviews for Wordscapes. Retrieved May 25, 2024 from https://www.appgamer.com/wordscapes/reviews/
4. Anon. Looking for games like Wordle? these fun word puzzles are just as addictive. Retrieved May 30, 2024 from https://www.gazetteleader.com/sponsored-content/looking-for-games-like-wordle-these-fun-word-puzzles-are-just-as-addictive-8426553
5. Anon. Paced Auditory Serial Addition Test (PASAT). Retrieved June 5, 2024 from https://www.nationalmssociety.org/for-professionals/for-researchers/researcher-resources/research-tools/clinical-study-measures/pasat#:~:text=Description,as%20well%20as%20calculation%20ability.
6. Anon.Retrieved June 15, 2024 from https://www.das.nh.gov/wellness/Docs%5CPercieved%20Stress%20Scale.pdf
7. Anon.Retrieved June 12, 2024 from https://random-word-api.herokuapp.com/home
8. Anon.Retrieved June 12, 2024 from https://dictionaryapi.dev/
9. Brooke Kato. 2024. Wordle driving you mad? why the New York Times Game is a problem for mental health, therapists say. (April 2024). Retrieved May 22, 2024 from https://nypost.com/2024/04/18/lifestyle/wordle-driving-you-mad-how-the-popular-game-hurts-mental-health/
10. Canva.Retrieved June 14, 2024 from https://www.canva.com/colors/color-wheel/
11. Coursera Staff. 2023. What is a wireframe? + how to create one. (November 2023). Retrieved June 14, 2024 from https://www.coursera.org/articles/wireframe
12. Goh Yan Han. 2024. Research to look at surge in youth mental health issues; causes could include social media use. (February 2024). Retrieved June 7, 2024 from https://www.straitstimes.com/singapore/politics/research-to-look-at-surge-in-youth-mental-health-issues-causes-could-include-social-media-use
13. Hamed Aliyari, Hedayat Sahraei, Sahar Golabi, Masoomeh Kazemi, Mohammad Reza Daliri, and Behrouz Minaei-Bidgoli. 2021. The effect of brain teaser games on the attention of players based on hormonal and brain signals changes. (September 2021). Retrieved May 22, 2024 from https://doi.org/10.32598/bcn.2021.724.9
14. Jasper van der Hoek. 2024. The 5 stages of the Agile Software Development Lifecycle. (April 2024). Retrieved June 3, 2024 from https://www.mendix.com/blog/agile-software-development-lifecycle-stages/
15. Jaymar Cabebe. 2014. Spelltower Review: SpellTower. (March 2014). Retrieved June 3, 2024 from https://www.cnet.com/reviews/spelltower-review/
16. Justin Davis. 2012. Six underrated iphone word games. (March 2012). Retrieved June 5, 2024 from https://www.ign.com/articles/2012/03/29/six-underrated-iphone-word-games
17. Moonee. Games publisher | color theory - tips for Game development | blog Moonee. Retrieved June 6, 2024 from https://www.moonee.io/blog/coler-theory-tips-for-game-development

18. Nathan Thomas. 2019. How to use the system usability scale (SUS) to evaluate the usability of your website. (September 2019). Retrieved June 11, 2024 from https://usabilitygeek.com/how-to-use-the-system-usability-scale-sus-to-evaluate-the-usability-of-your-website/
19. Newport Institute Staff. Brain rot: The impact on Young Adult Mental Health. Retrieved May 21, 2024 from https://www.newportinstitute.com/resources/co-occurring-disorders/brain-rot/
20. OpiumPulses. 2021. Wordscapes Review (mobile): Opium pulses. (June 2021). Retrieved May 21, 2024 from https://www.opiumpulses.com/article/458/wordscapes-review#google_vignette
21. Sue Howe. 2024. Social Media Statistics in Singapore [Updated 2024] Meltwater (May 2024). Retrieved June 13, 2024 from https://www.meltwater.com/en/blog/social-media-statistics-singapore.
22. Tech Desk. 2021. Wordle: All about the viral game twitter is going bonkers over. (December 2021). Retrieved May 29, 2024 from https://indianexpress.com/article/technology/social/wordle-all-about-the-viral-game-twitter-is-going-bonkers-over-7682968/
23. Toni Vujevic. 2023. The Mobile App Architecture Guide. (March 2023). Retrieved June 6, 2024 from https://decode.agency/article/mobile-app-architecture/
24. Venngage. Accessible color palette generator: WCAG compliant. Retrieved June 6, 2024 from https://venngage.com/tools/accessible-color-palette-generator
25. Aadil Mallick. 2023. How to use expo snacks in react native: LFT #2. (January 2023). Retrieved September 9, 2024 from https://medium.com/@waadlingaadil/how-to-use-expo-snacks-in-react-native-lft-2-653853c9f503
26. React Native. 2024a. 🚧 asyncstorage · REACT native. (August 2024). Retrieved September 9, 2024 from https://reactnative.dev/docs/asyncstorage
27. Anon. 2024b. How to write test cases (with format & example). (May 2024). Retrieved September 9, 2024 from https://www.browserstack.com/guide/how-to-write-test-cases
28. Anon. Collegiate dictionary API: Merriam-Webster Dictionary API. Retrieved September 9, 2024a from https://dictionaryapi.com/products/api-collegiate-dictionary
29. Anon. English Letter Frequency (based on a sample of 40,000 words). Retrieved September 9, 2024b from https://pi.math.cornell.edu/~mec/2003-2004/cryptography/subs/frequencies.html
30. Anon. Find any sound you like. Retrieved September 9, 2024c from https://freesound.org/
31. Anon. Let's build from here. Retrieved September 9, 2024d from https://github.com/
32. React Native. React native · learn once, write anywhere. Retrieved September 9, 2024e from https://reactnative.dev/
33. Wordnik. "sign up for a wordnik API key!" Retrieved September 9, 2024f from https://developer.wordnik.com/
34. American Psychology Association. Social media brings benefits and risks to teens. psychology can help identify a path forward. Retrieved September 9, 2024g from https://www.apa.org/monitor/2023/09/protecting-teens-on-social-media
35. Stack Navigator. Stack navigator: React navigation. Retrieved September 9, 2024h from https://reactnavigation.org/docs/stack-navigator/
36. Pixabay. Retrieved September 9, 2024i from https://pixabay.com/